

A Modular Systems-Biology Framework for nsPEF-Enabled Extracellular Vesicle Engineering

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Abstract

Nanosecond pulsed electric fields (nsPEF) can stimulate extracellular vesicle (EV) release through coupled membrane electrodynamics, organelle perturbation, Ca^{2+} redistribution, membrane remodeling, repair, vesicle biogenesis, and stress-dependent cargo sorting. However, these processes span nanosecond electrical events, millisecond-to-second Ca^{2+} and organelle signaling, and minute-to-hour EV release and cargo phenotypes, making whole-cell simulation impractical for early model development. Here we define a modular, cell-type-adaptable systems-biology framework for nsPEF-enabled EV engineering. The framework organizes the mechanism into linked layers: pulse delivery and dosimetry; plasma membrane and organelle electrodynamics; ion transport, Ca^{2+} mobilization, ROS, and bioenergetics; Ca^{2+} -dependent remodeling and repair; EV biogenesis and subtype release; cargo sorting, EV composition, and potency; injury, debris, and quality gating; and manufacturing, isolation, and quality-control interfaces. Cell-state and disease-state modifiers act as cross-cutting parameter layers that adapt the shared model to cancer cells, endothelial cells, immune cells, stromal cells, and other EV-producing systems. This structure provides a transparent framework for connecting pulse parameters to EV yield, subtype distribution, cargo state, functional potency, purity, and cell viability.

Keywords

nanosecond pulsed electric fields; electro-exocytosis; extracellular vesicles; EV engineering; systems biology; compartmental modeling; calcium signaling; electroporation; cargo modulation; biomanufacturing

Abbreviations and Nomenclature

Table 1 summarizes the core biological abbreviations used throughout the manuscript. Variable symbols and parameter notation are listed separately in Appendix Table A17.

Introduction

Nanosecond pulsed electric fields (nsPEF) are ultrashort, high-intensity electric pulses capable of perturbing plasma membranes and intracellular membrane-bound organelles. Unlike conventional electroporation, which typically uses microsecond-to-millisecond pulses that primarily charge the plasma membrane, nsPEF can access intracellular membranes because the pulse duration is comparable to, or shorter than, relevant membrane-charging times^{9,99,129}. This physical distinction allows nsPEF to trigger rapid Ca^{2+} mobilization, organelle stress responses, phospholipid redistribution, cytoskeletal remodeling, and regulated or stress-associated vesicle release^{8,68,83}. nsPEF have emerged as a bioelectric modality in cancer and other biomedical applications, and recent experimental evidence suggests that they can also stimulate robust extracellular vesicle (EV) release, a process referred to here as *electro-exocytosis*⁶⁰.

EVs are heterogeneous membrane-bound particles involved in intercellular communication, immune modulation, vascular remodeling, tissue repair, and disease progression. In accordance with MISEV2023 nomenclature principles, this manuscript uses the umbrella term EV when vesicle origin has not been experimentally resolved¹³⁸.

Mechanistic terms such as exosome, microvesicle, and apoptotic body are used when referring to biogenesis-specific model pathways. This distinction is important for modeling because measured EV populations often reflect mixtures of small EVs, medium/large EVs, apoptotic bodies, damaged organellar fragments, protein aggregates, and other co-isolated materials. A useful nsPEF-EV model must therefore predict not only total particle release, but also subtype weighting, cargo state, functional potency, purity, and the boundary between controlled EV release and injury-associated debris.

At the cellular level, nsPEF stimulation initiates a cascade that bridges membrane electrodynamics with vesicle biogenesis. Rapid transmembrane-potential changes and nanopore

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Table 1. Core abbreviations used throughout the manuscript.

Abbr.	Definition
nsPEF	nanosecond pulsed electric field
EV	extracellular vesicle
sEV	small extracellular vesicle
m/IEV	medium/large extracellular vesicle
AB	apoptotic body
MVB	multivesicular body
ILV	intraluminal vesicle
ER	endoplasmic reticulum
ROS	reactive oxygen species
ATP	adenosine triphosphate
PS	phosphatidylserine
ESCRT	endosomal sorting complex required for transport
QC	quality control

formation enable ionic fluxes, most notably Ca^{2+} influx and redistribution, with patch-clamp and imaging studies showing minute-scale permeability recovery and Ca^{2+} thresholds that depend on pulse width and cell type^{74,80,81,99}. Intracellular membranes such as the ER, mitochondria, endosomes, and MVBs may also be perturbed, creating routes for store release, mitochondrial stress, ROS generation, trafficking changes, and cargo-sorting modulation^{8,71,100}. Local Ca^{2+} signals then activate effectors including annexins, calpain, scramblases, phospholipases, synaptotagmin-like sensors, and SNARE-associated fusion machinery. These signals drive phosphatidylserine externalization, cytoskeletal reorganization, membrane curvature, MVB docking and fusion, microvesicle budding, and membrane repair^{12,68,69,140}.

From a computational systems-biology perspective, nsPEF-induced EV engineering is best formulated as a modular pathway-level perturbation rather than as a complete whole-cell simulation. In this framework, the electrical pulse is represented as a controllable external input that modifies plasma membrane and organelle permeability, producing time-dependent Ca^{2+} , ROS, ATP, repair, remodeling, trafficking, and injury states. These upstream variables are then mapped to EV release rates, subtype weighting, cargo composition, functional potency, and quality-control outputs. Such a structure preserves mechanistic interpretability while allowing the same core model to be instantiated in cancer cells, immune cells, endothelial cells, mesenchymal stromal cells, or isolated EVs.

The present manuscript therefore focuses on the modeling problem: how to connect an externally defined nsPEF exposure to membrane and organelle perturbation, Ca^{2+} -dependent remodeling, repair, EV biogenesis, cargo modulation, injury-quality gating, and manufacturing-facing EV outputs through a transparent and literature-curatable computational framework.

Methods

We formulated nsPEF-enabled EV engineering as a modular systems-biology framework that links pulse exposure to membrane and organelle perturbation, Ca^{2+} signaling, membrane remodeling, repair, vesicle release, cargo state, injury-quality gating, and manufacturing-facing outputs. Each biological process is represented as a model module with defined inputs, state variables, outputs, and

candidate mathematical forms. Paired appendix tables **A1–A16** summarize biological evidence and computational implementation options.

- (i) Define the minimal set of biological and engineering layers required to connect nsPEF exposure to EV yield, subtype distribution, cargo state, potency, purity, and viability.
- (ii) Assign each layer a computational representation appropriate to its dominant timescale, state variables, measurable outputs, and available modeling literature.
- (iii) Separate biological evidence curation from computational model curation for each module.
- (iv) Distinguish cell-based electro-exocytosis from direct post-isolation EV engineering.
- (v) Represent cell type, disease state, activation state, and population heterogeneity as cross-cutting modifiers rather than as a downstream pathway.

In vitro particle-size measurements

Size-resolved extracellular particle measurements were obtained from CD4^+ cell preparations nominally containing 5×10^6 cells per experimental batch. The supplied export contained conditions labeled as one pulse at 20 kV, three pulses at 40 kV, and five pulses at 40 kV, sampled at 0.5, 1, and 3 h after exposure. Particle concentrations were rebinned into 15 common 20-nm bands spanning 80–380 nm before calculating arithmetic means and sample standard deviations. The analysis included 26 stimulated histograms, with three repeated measurements per condition and time except for the three-pulse, 40-kV condition at 1 h ($n = 2$). Three undated, cell-containing sham histograms were averaged to provide a provisional initial extracellular distribution. Three separately labeled, undated “sham media” histograms were excluded because their relationship to the cell-containing controls was unresolved. The supplied metadata did not establish the independence of repeated measurements; consequently, they are not described here as biological replicates. Complete culture, exposure, collection, sample-processing, and particle-analysis procedures remain to be confirmed by the experimental study team.

Size-resolved observation and optimization pipeline

The intracellular model was retained without modification and supplied cumulative per-cell release trajectories for the sEV, m/IEV, and AB pathways. The inter-layer bridge described in the Appendix converted increments in those trajectories to a latent extracellular particle distribution over 40–2000 nm. For pathway q , release was distributed through a nonnegative lognormal kernel whose median could respond to the matched upstream cell-state signal,

$$d_{50,q}(t) = d_{50,q}^0 \exp[\gamma_q S_q(t)], \quad (1)$$

where $d_{50,q}^0$ is the baseline median diameter, γ_q is the state-coupling coefficient, and $S_q(t)$ is the difference between the corresponding exposure trajectory and a simulated near-zero-field sham proxy. This simulated reference is distinct from the experimental sham histograms used for extracellular initialization. The formulation preserves mechanistic pathway

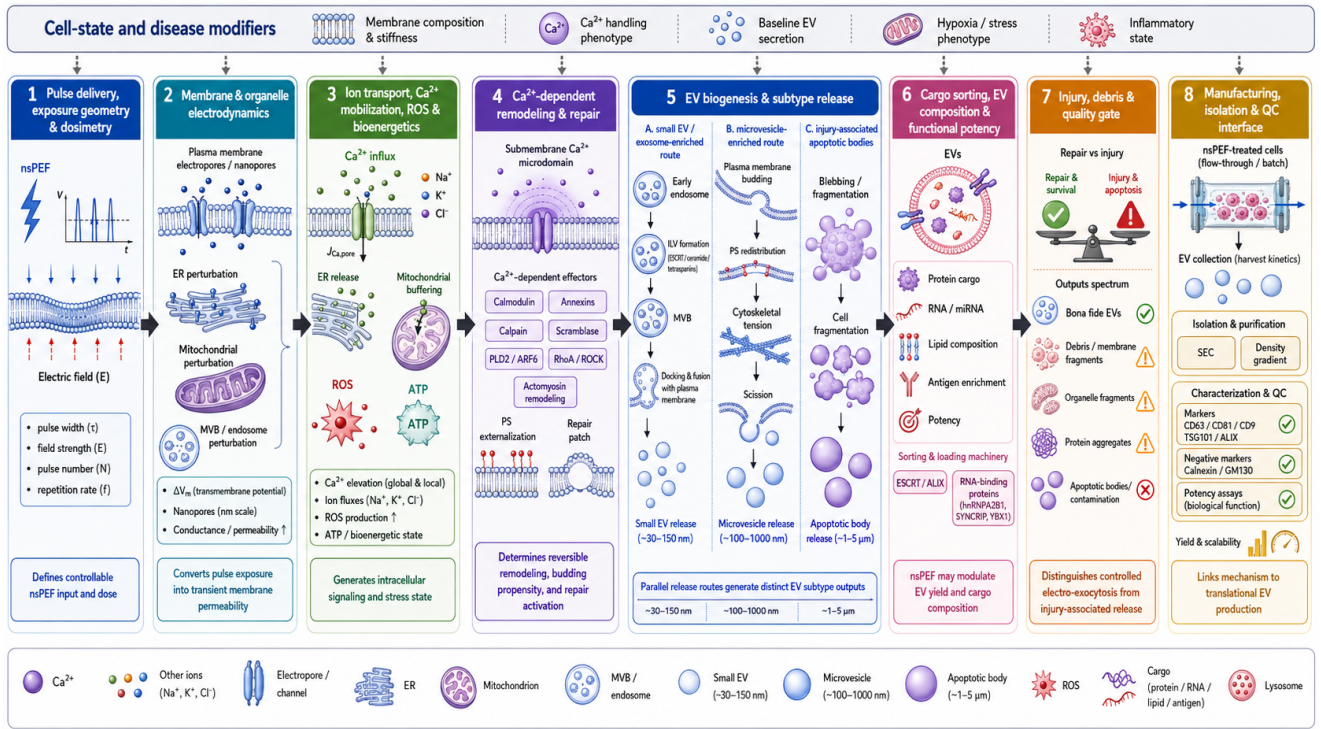


Figure 1. Integrated mechanisms linking nanosecond pulsed electric field stimulation to extracellular vesicle (EV) engineering. Nanosecond pulsed electric fields induce membrane and organelle perturbation, Ca²⁺ redistribution, phosphatidylserine externalization, cytoskeletal remodeling, and repair activation. These processes can promote EV release through endosomal/MVB fusion, plasma membrane budding, and injury-associated vesiculation. In an EV-engineering framework, model outputs include EV yield, size/subtype distribution, cargo composition, functional potency, purity, and cell viability.

Abbreviations: EV, extracellular vesicle; ILV, intraluminal vesicle; MVB, multivesicular body; ESCRT, endosomal sorting complex required for transport; VGCC, voltage-gated calcium channel; PLD2, phospholipase D2; ARF6, ADP-ribosylation factor 6; SNARE, soluble N-ethylmaleimide-sensitive factor attachment protein receptor; LAMP1, lysosome-associated membrane protein 1; HIF-1α, hypoxia-inducible factor 1-α; IP₃R, inositol 1,4,5-trisphosphate receptor; PS, phosphatidylserine.

identities while allowing their operational size distributions to overlap, as required for particle-based observations¹³⁸.

The extracellular concentration C_b in latent size band b was propagated with pathway release input and a smooth size-dependent first-order loss,

$$\frac{dC_b}{dt} = \rho_v(t) \sum_q a_q K_{qb}(t) r_q(t) - \frac{\ln 2}{t_{1/2,150}} \left(\frac{d_b}{150 \text{ nm}} \right)^\eta C_b, \quad (2)$$

where $\rho_v(t)$ is viable producer-cell density, a_q converts pathway-model units to particle equivalents, K_{qb} is the pathway-to-size kernel mass, $r_q(t)$ is the pathway release rate, $t_{1/2,150}$ is the effective extracellular half-life at 150 nm, and η specifies the size dependence of loss. Initial extracellular particles and newly released particles experienced the same loss law. An explicit observation matrix then mapped latent stocks to the measured diameter bands. Because instrument-response and recovery calibrations were unavailable, the preliminary operator was identity-like over 80–380 nm, with no broadening, unit recovery, and zero background¹²⁵.

The experimental-to-computational bridge used 5×10^6 initial cells and a provisional 5-mL medium volume, equivalent to 10^6 initial cells mL⁻¹. No measured viability trajectory was available; the bridge-level viability normalization,

recovery, and dilution factors were provisionally set to one, while $\rho_v(t)$ retained the simulation's time-varying viable-producer fraction. These quantities are interface assumptions rather than confirmed experimental metadata, and fitted source-conversion factors absorb their uncertainty. Similarly, the computational scenarios assumed 60-ns pulses at 1 Hz and provisionally interpreted the numerical 20- and 40-kV source labels as kV cm⁻¹; no inference about the delivered field should be made until exposure metadata are confirmed.

Calibration separated the total concentration and normalized size composition. For condition c , time t , and measured band b , residuals were

$$r_{ct}^{\text{total}} = \ln \left(\frac{\hat{T}_{ct}}{T_{ct}} \right), \quad r_{ctb}^{\text{size}} = \sqrt{\hat{f}_{ctb}} - \sqrt{f_{ctb}}, \quad (3)$$

where T is total concentration over 80–380 nm and f is the corresponding size-bin fraction. The square-root composition residual accommodates empty bins without pseudocounts. Sample standard deviations were retained for descriptive plotting but were not used as inverse-variance weights because only two or three repeated measurements were available and bands from each histogram were correlated. Weak penalties (weight 0.25) regularized pathway medians and widths, the size-loss slope, state couplings, and the optional exposure-condition adapter toward their initial values.

Three nested observation-model variants were compared: time-invariant fitted size kernels, state-conditioned kernels, and state-conditioned kernels with a shared empirical dose-response gain. For exposure condition c , the gain was

$$g_c = \exp[\beta_1(D_c - 0.72) + \beta_2(D_c - 0.72)^2], \quad (4)$$

where D_c is the model-derived dimensionless dose index and 0.72 is a fixed centering value; g_c multiplied all three pathway release trajectories. Each variant was fitted by bounded nonlinear least squares from 24 initial vectors—one canonical vector and 23 seeded Latin-hypercube points spanning the configured bounds—with at most 100 function evaluations per start. The reported selection used a descriptive composite penalized score. Because the residual vector combines total and compositional errors rather than deriving from a complete sampling likelihood, this score is a heuristic ranking and not a formal information criterion. Fit quality was summarized by $\log_{10}$ total-concentration RMSE and its fold equivalent, percentage error, Hellinger and Wasserstein distances between normalized size distributions, and error in distribution mean diameter.

Results

The resulting framework is a cell-type-adaptable module map rather than a single parameterized model. Table 2 provides the concise navigation map. Table 3 defines the mathematical modeling strategy and module-interface logic. The appendix tables then provide the biological and computational evidence-curation structure needed for future model implementation and parameterization.

Preliminary results

To demonstrate how the current implementation behaves before formal calibration, we generated a set of targeted example simulations aligned to the first five layers of the framework. These examples are not intended as validation studies. Rather, they show that the selected mathematical forms already produce interpretable regime structure: dosimetry responds to conductivity and pulse-train timing, electrodynamics responds to pulse width and cell size, compartmental ion-transport equations generate coupled Ca^{2+} /ROS/ATP trajectories, remodeling rules separate buffering from repair-amplification effects, and the EV-release pool model distinguishes productive secretory windows from injury-dominant vesiculation.

The dosimetry example therefore illustrates the value of a selectable Layer 1 representation rather than a single absorbed-energy scalar. The legacy, adiabatic, and lumped-retention variants produce similar exposure ordering but different thermal interpretations, which is exactly the modeling role intended in Table A2. The electrodynamics example then shows the consequence of the Layer 2 reduced charging model: pulse width is not merely a protocol label, but an explicit determinant of how close the membrane gets to its Schwan-limit voltage during each pulse. This matters downstream because the same voltage state drives the permeability and pore-density surrogates that feed the Ca^{2+} and organelle modules.

The Layer 3 example is especially useful because it shows how the compartmental flux-balance formulation behaves

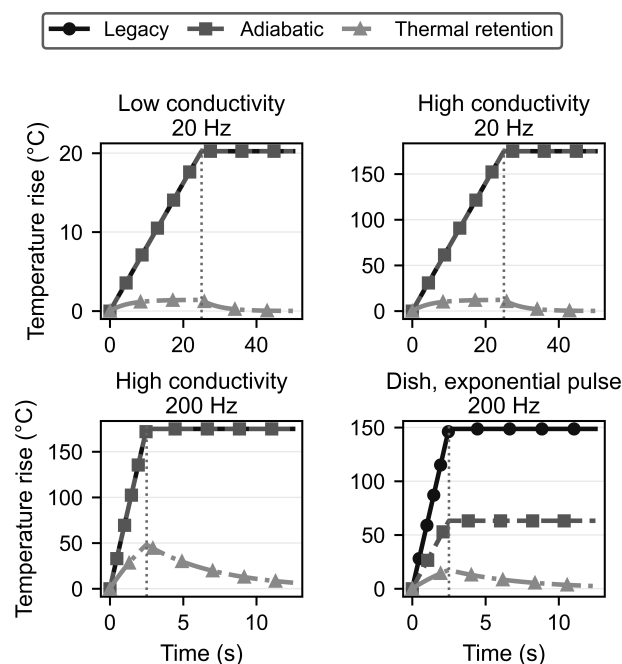


Figure 2. Preliminary Layer 1–2 example behavior. The dosimetry comparison shows how the lumped thermal framework separates absorbed electrical dose from retained heating: low-conductivity conditions remain weakly heated, while high-conductivity and high-repetition-rate pulse trains accumulate larger temperature rises even under similar nominal amplitudes.

under qualitatively different assumptions. Increasing pulse severity amplifies cytosolic Ca^{2+} , ROS, and downstream stress, whereas calcium-limited conditions blunt the Ca^{2+} response even when the same electrodynamic trigger is applied. Slowing ion recovery extends osmotic imbalance, which in turn sustains damage pressure despite a similar initial poration event. These behaviors are consistent with the purpose of Table A6: the module is meant to translate a brief electroporation event into longer-lived signaling, organelle, and metabolic trajectories.

The Layer 4 example then shows why the remodeling layer is not redundant with Layer 3. In the present Hill-activation and repair-state framework, buffered microdomains reduce submembrane Ca^{2+} amplification and therefore suppress phosphatidylserine exposure and downstream remodeling. Calpain inhibition preferentially reduces actin disruption without removing the upstream Ca^{2+} transient itself, while stronger annexin/lysosomal support preserves a larger resealing state over time. This is the intended function of Table A8: to resolve whether a given Ca^{2+} /stress history is interpreted as recoverable remodeling or escalating injury.

The Layer 5 example is particularly important because it demonstrates the value of the explicit MVB/ILV/docking/fusion/budding state structure. Rab/ESCRT limitation reduces small-EV release without requiring a generic global suppression of all vesiculation, whereas a ceramide-favored secretory bias increases the release-competent MVB route by reducing lysosomal routing. In contrast, the injury-shift scenario sharply increases apoptotic-body production and collapses

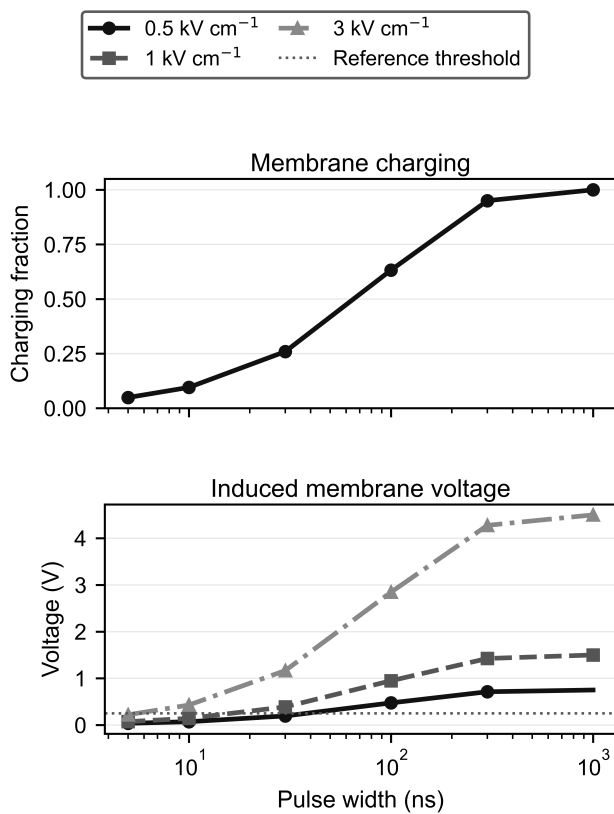


Figure 3. Preliminary Layer 1–2 example behavior. The reduced electrodynamics model shows first-order membrane charging toward the Schwan limit, so longer pulses and larger cells reach higher induced membrane voltages and cross a permeability-relevant threshold more readily than ultrashort or smaller-cell cases.

viability. This is exactly the interpretive gain intended by the Table A10 framework: subtype release becomes a competition between secretory trafficking, membrane budding, and apoptotic disassembly rather than a single EV-rate output.

The integrated storyboard provides a compact narrative view of the framework. In the mild reversible window, the model predicts threshold-level charging, modest temperature rise, limited Ca²⁺ amplification, and predominantly repair-compatible remodeling with low apoptotic output. In the productive secretory window, larger Ca²⁺ loading and stronger repair support coexist with a more favorable secretory routing bias, increasing small-EV output without a catastrophic loss of viability. In the injury-dominant window, the same layered structure yields a different interpretation: stronger charging and thermal load amplify Ca²⁺ and ROS, repair remains incomplete, and apoptotic vesiculation becomes a dominant endpoint. This row-wise transition is encouraging because it shows that the current framework already produces coherent scenario-level behavior while remaining modular enough for later biological calibration.

The Layer 6–8 comparison illustrates why the implementation now carries cargo, quality, and manufacturing states beyond total particle release. Productive and

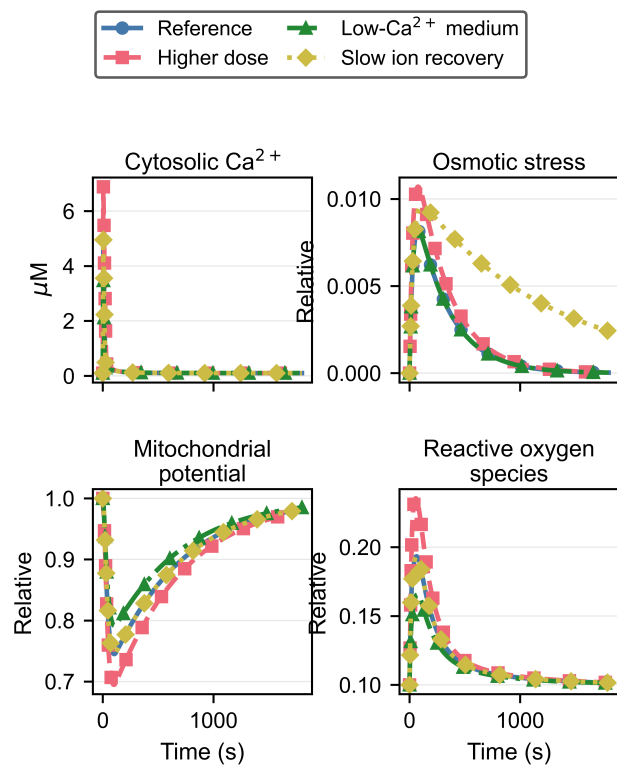


Figure 4. Preliminary Layer 3–4 example behavior. The compartmental ion-transport/bioenergetics example compares baseline, higher-dose, calcium-limited, and slow-recovery settings; the chosen ODE framework converts the electroporation state into different Ca²⁺ transients, osmotic stress histories, mitochondrial stress, and ROS accumulation.

direct-loading scenarios can yield similar process objectives through different mechanisms—endogenous cargo sorting versus permeability-assisted loading—whereas the injury-dominant scenario shows that a high particle output is not automatically a useful EV product. This behavior follows the report’s computational tables: subtype-weighted cargo and stress sorting feed potency, damage and contaminant mixtures determine the quality gate, and recovery, purity, batch consistency, and potency jointly determine the manufacturing objective.

In vitro size-resolved particle kinetics

The extracellular particle measurements showed condition-dependent and nonmonotonic redistribution across the common 80–380-nm analysis window (Figure 9). Although whole-distribution mean diameters remained between approximately 134 and 147 nm, individual 20-nm bands exhibited heterogeneous temporal changes, and adjacent bands could change in different directions between the same two harvest times. Total concentration or one mean-diameter statistic therefore cannot fully represent the observed response. These descriptive profiles support the inclusion of extracellular loss, but three measured times do not identify a continuous kinetic trajectory. Within-group concentration totals were much less variable than the shapes of the corresponding size distributions, which may reflect source-file normalization or a shared condition-level concentration;

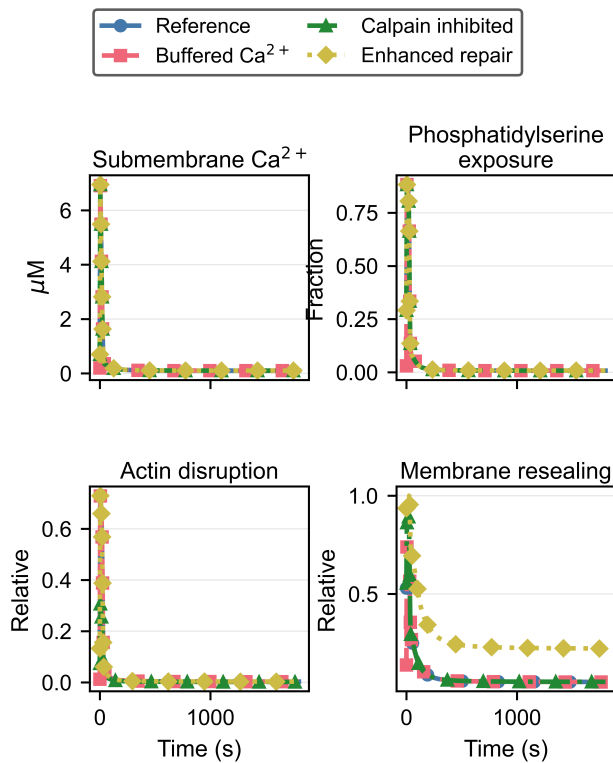


Figure 5. Preliminary Layer 3–4 example behavior. The remodeling/repair example shows how the reduced Ca^{2+} -sensor module separates phosphatidylserine exposure, actin disruption, and resealing competence across buffered, calpain-inhibited, and annexin/lysosome-supported regimes.

the reported variability is therefore descriptive pending clarification of the acquisition and export workflow.

Preliminary calibration of the size-resolved extracellular model

The state-conditioned model with the empirical dose-response adapter achieved the lowest descriptive composite score. Across nine condition–time targets, its total-concentration $\log_{10}$ RMSE was 0.1042, corresponding to a typical multiplicative error of 1.27-fold. Median absolute percentage error was 18.0%, and median symmetric absolute percentage error was 19.7%. For size composition, the mean Hellinger distance was 0.0857, the mean Wasserstein distance was 5.00 nm, and the mean absolute error in distribution mean diameter was 4.66 nm. Descriptively, 22.2% of total-concentration targets and 56.3% of size-band predictions fell within the observed mean $\pm$ sample standard deviation. All of these statistics are in-sample diagnostics rather than estimates of predictive performance.

The time-invariant-kernel and state-conditioned variants produced essentially identical total-concentration errors ($\log_{10}$ RMSE, 0.1879 for both) and mean Hellinger distances (0.0974 and 0.0973, respectively). Adding the empirical condition adapter reduced the total error to 0.1042 and the mean Hellinger distance to 0.0857. Thus, state-dependent shifts in kernel median did not explain the observed separation among exposure conditions, whereas an empirical pulse-condition correction improved the fit. This adapter is a diagnostic indication of missing condition dependence

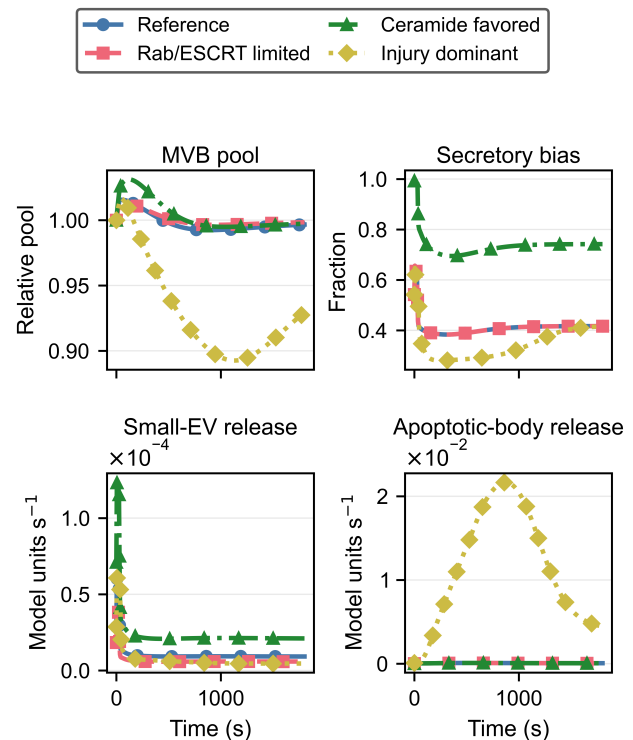


Figure 6. Preliminary Layer 5 example behavior. The EV-biogenesis comparison shows that the pool-balance release model distinguishes Rab/ESCRT-limited, ceramide-favored secretory, and injury-shifted regimes. In this framework, MVB pool size and secretory bias respond differently from final apoptotic-body output, allowing the model to separate productive small-EV routing from high-stress vesiculation.

Abbreviations: EV, extracellular vesicle; MVB, multivesicular body; ESCRT, endosomal sorting complex required for transport.

in the current upstream model, not a validated constitutive mechanism.

The selected fit reproduced the dominant concentration ridge near 100–160 nm and part of its redistribution across the three harvest times (Figure 9). The remaining error was structured rather than uniformly random (Figure 10): the largest relative discrepancies occurred in low-concentration tail bands and around condition-specific peaks. The error map is restricted to the measured 0.5-, 1-, and 3-h rows and does not imply values at intermediate times.

The selected fit gave an effective extracellular half-life of 0.680 h at 150 nm and a size-loss exponent of -0.0576 , indicating only weak size dependence in the inferred loss. The fitted sEV and m/IEV kernel medians were 121.5 and 201.2 nm, with geometric standard deviations of 1.229 and 1.424, respectively. The AB source scale approached its lower search bound, while its median and width remained near the nominal 420 nm and 1.60 values. The AB contribution is therefore not identifiable from the 80–380-nm observations.

All 24 optimization starts converged, and 18 endpoints were within 1% of the best regularized objective. Among these near-optimal endpoints, the effective half-life ranged from 0.658 to 0.680 h, the loss exponent from -0.0576 to -0.0344 , the sEV median from 121.485 to 121.655 nm, and the m/IEV median from 200.612 to 202.461 nm. In contrast,

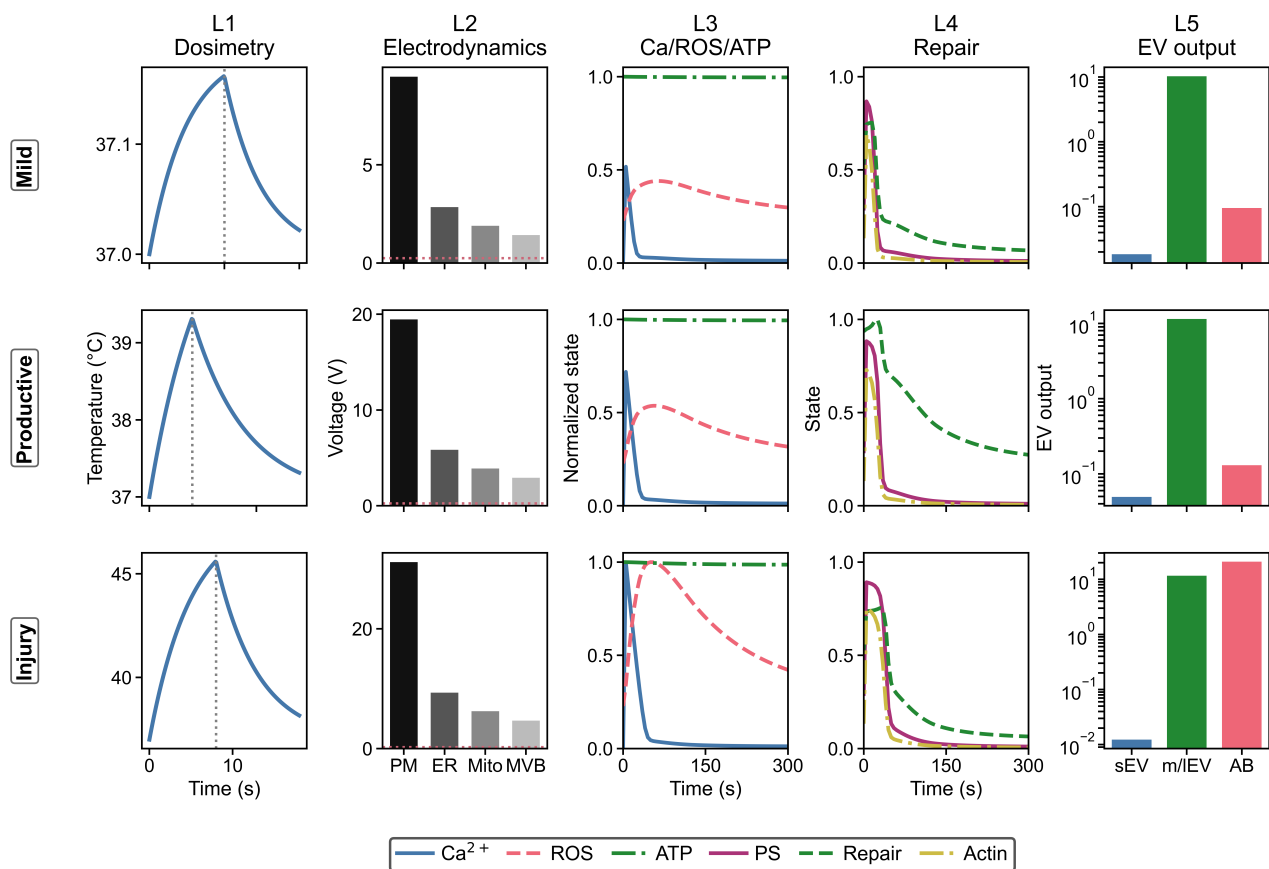


Figure 7. Integrated preliminary storyboard across Layers 1–5. Each row represents one illustrative scenario and each column summarizes one model layer from dosimetry through EV output. The top row shows a mild reversible window, the middle row a more productive secretory window, and the bottom row an injury-dominant window. The figure makes the multiscale logic of the framework visible: stronger Layer 1 dose and Layer 2 charging drive larger Layer 3 Ca^{2+} /ROS perturbations, which then reshape Layer 4 repair states and finally alter Layer 5 subtype weighting and viability.

Abbreviations: Ca^{2+} , calcium ion; ROS, reactive oxygen species; ATP, adenosine triphosphate; EV, extracellular vesicle; sEV, small extracellular vesicle; AB, apoptotic body.

the AB source conversion ranged from 0.0103 to 46.1 particles per pathway-model unit. These ranges diagnose numerical optimization stability under the configured starts and bounds; they are not confidence intervals, posterior distributions, or estimates of biological variability. More densely sampled time courses, confirmed sham harvest times, instrument-response calibration, and independent validation conditions will be required before interpreting the fitted values as constitutive biological parameters.

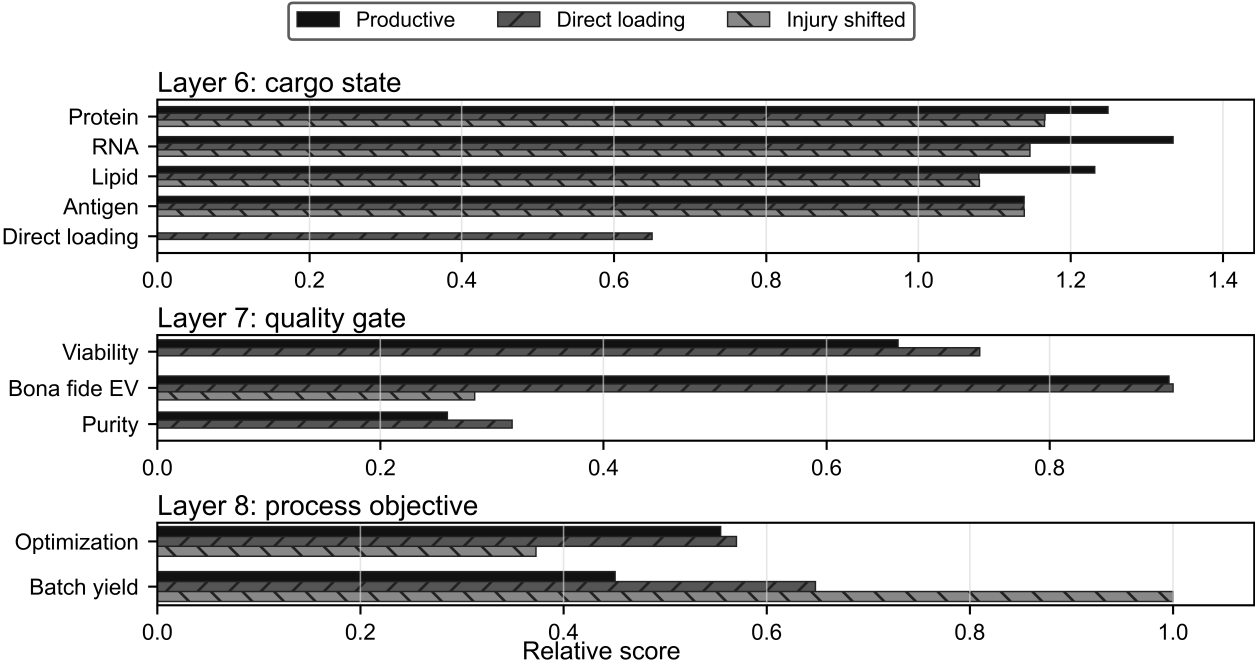


Figure 8. Preliminary Layer 6–8 downstream engineering behavior. The downstream comparison extends the preliminary examples from vesicle release into product interpretation. A productive cargo window increases protein, RNA, lipid, and antigen cargo scores through secretory and ESCRT-linked terms; a direct-loading mode adds a permeability-driven cargo term while retaining acceptable viability; and an injury-dominant condition may still produce particles but fails the quality gate because apoptosis, necrosis, and contaminant fractions reduce bona fide EV purity and process objective.

Abbreviations: EV, extracellular vesicle; RNA, ribonucleic acid; ESCRT, endosomal sorting complex required for transport.

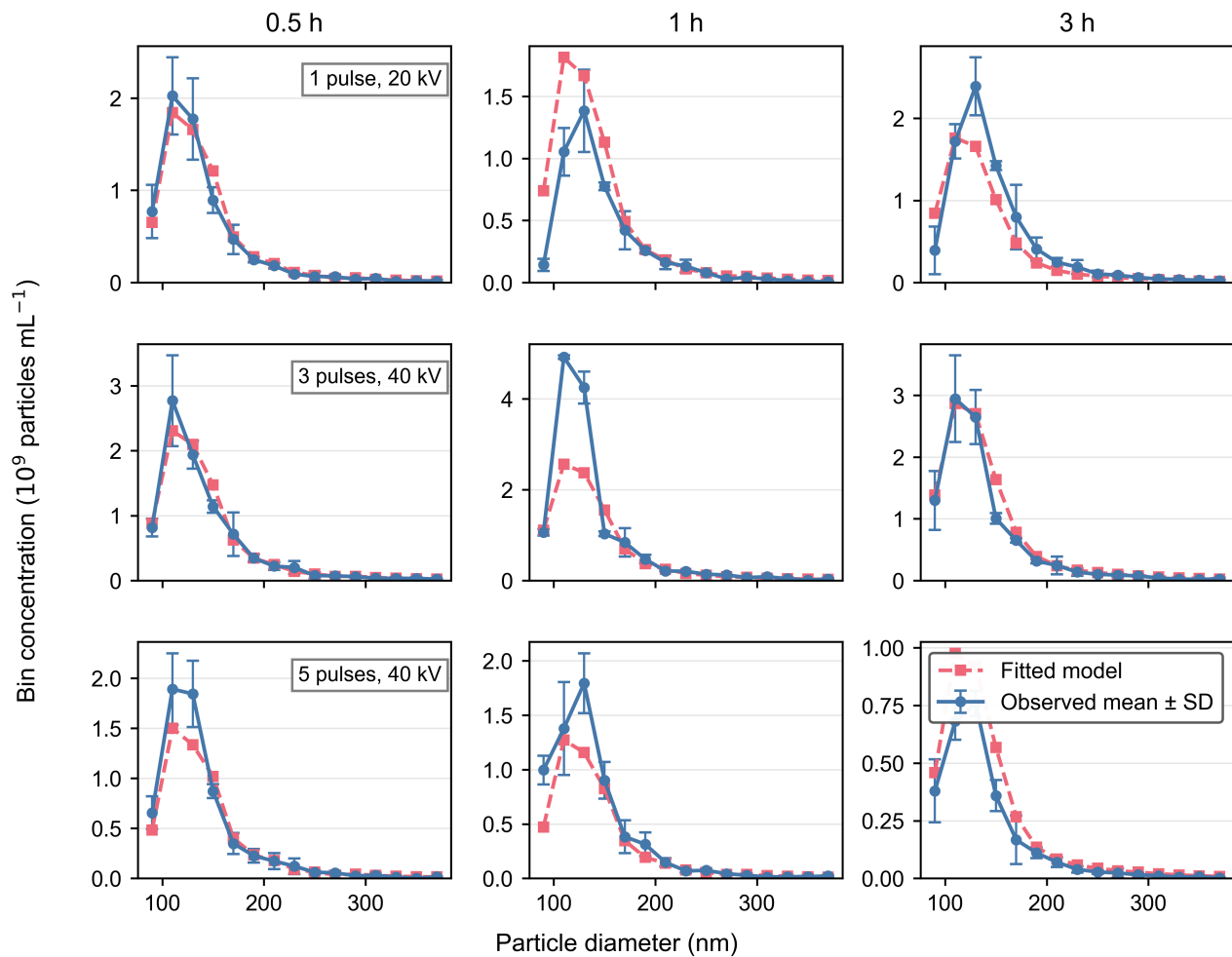


Figure 9. Size-resolved extracellular particle concentrations and selected fitted model. Observations were rebinned into 20-nm bands over 80–380 nm and are shown as mean $\pm$ sample standard deviation; fitted values are shown by dashed curves. Each panel represents one exposure condition and harvest time. There were three repeated measurements per condition and time except for the three-pulse, 40-kV condition at 1 h ($n = 2$). Diameter bands from a single histogram are correlated, and the independence of repeated measurements was not established by the supplied metadata.

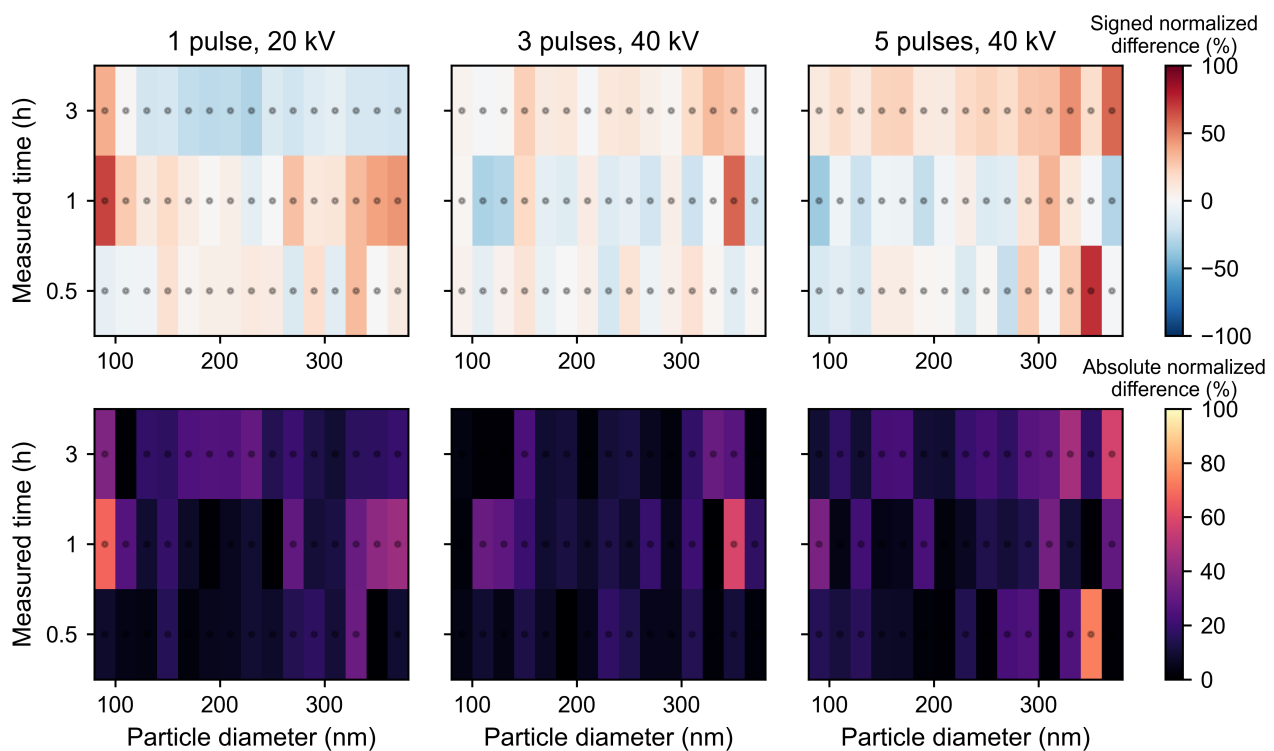


Figure 10. Size–time map of model error. Signed normalized difference was calculated as $100(\text{model} - \text{experiment})/(\text{model} + \text{experiment})$; the absolute map shows its absolute value. Blue indicates underprediction and red indicates overprediction in the signed maps. Each cell corresponds to an observed diameter band at 0.5, 1, or 3 h; no intermediate times are interpolated. Low-concentration tail bands can show large relative differences because an instrument detection-limit mask was unavailable. These maps are diagnostics and are distinct from the residuals used for optimization.

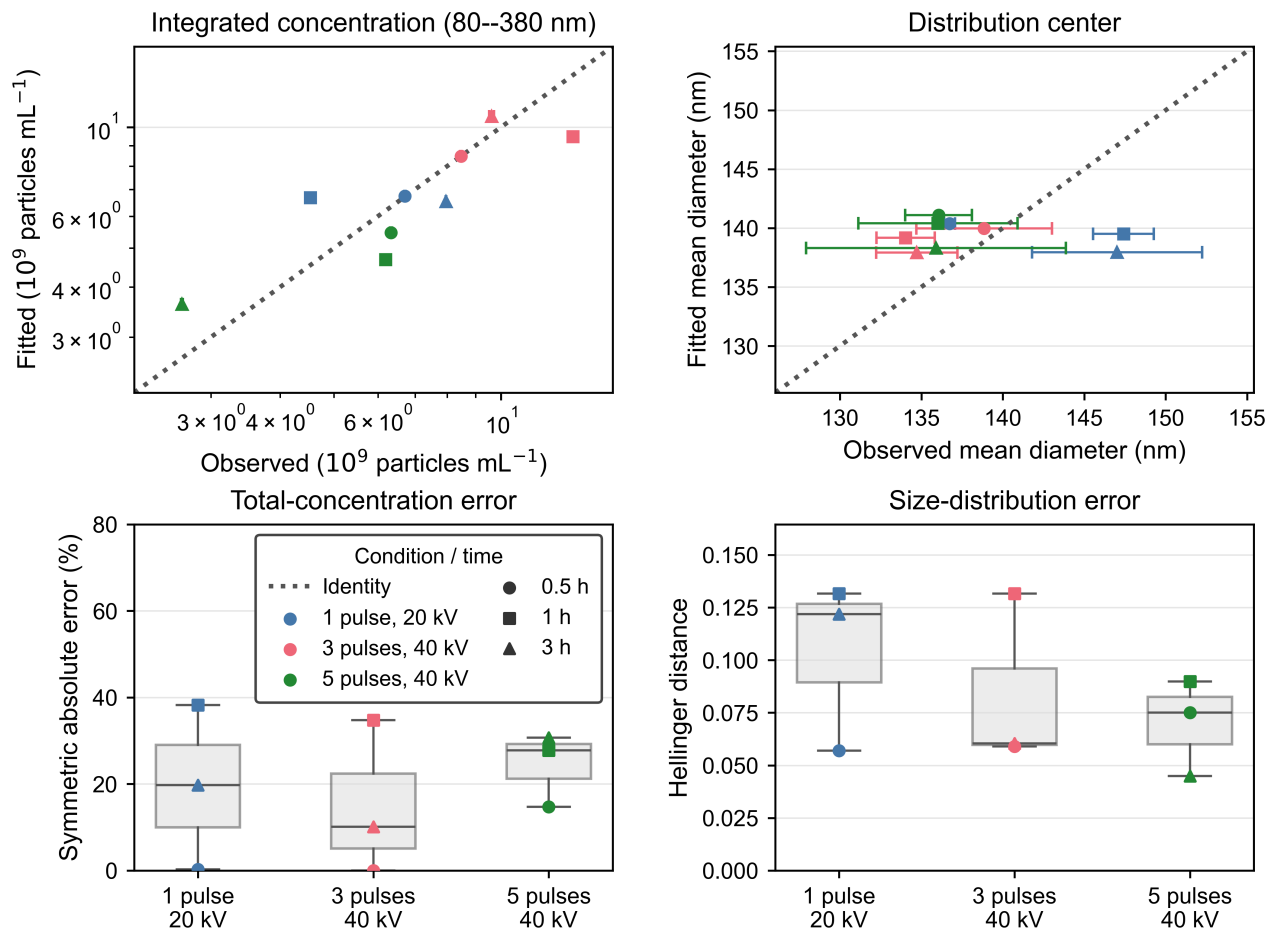


Figure 11. In-sample goodness-of-fit diagnostics for the selected size-resolved model. Upper panels compare fitted and observed integrated concentration and distribution mean diameter with identity lines; horizontal bars show observed sample standard deviation. Integrated concentration refers only to the common measured 80–380-nm window and should not be interpreted as total EV abundance. Mean-diameter values and their standard deviations were calculated from whole-histogram diameter means. Lower panels summarize symmetric absolute concentration error and Hellinger size-distribution distance across the three measured times. Boxes contain three condition-specific time points and are descriptive, not confidence intervals.

Discussion

The proposed structure shifts the conceptual endpoint of electro-exocytosis modeling from total EV release alone to nsPEF-enabled EV engineering. In this formulation, EV output is not a single scalar quantity but a multi-dimensional product state that includes particle yield, subtype weighting, cargo composition, potency, purity, and producer-cell viability. This distinction is essential because nsPEF can plausibly increase bona fide EV release, alter cargo sorting, stimulate repair-associated shedding, or generate injury-associated particles depending on pulse dose, cell state, and recovery kinetics. A systems model that does not separate these outputs risks conflating useful EV production with nonspecific stress debris.

A second key feature of the framework is the explicit treatment of organelle electrodynamics. nsPEF effects on the ER, mitochondria, endosomes, and MVBs provide a mechanistic route by which pulse parameters may influence not only Ca^{2+} dynamics but also exosome-biogenesis machinery, cargo sorting, lysosomal routing, and mitochondrial stress. These processes are difficult to resolve experimentally in a single study, but modular modeling allows them to be represented as testable submodels with defined inputs and outputs.

Finally, the framework distinguishes two related but separate applications. In cell-based electro-exocytosis, nsPEF is applied to producer cells to stimulate vesicle release and modify cargo through endogenous signaling and sorting pathways. In direct EV engineering, nsPEF is applied to isolated EVs to transiently permeabilize vesicle membranes for cargo loading or surface modification. These modes share membrane-electroporation principles but require different state variables, size scales, and quality-control outputs. Separating them prevents overextension of a single model while preserving a shared bioelectric modeling backbone.

Appendix

Appendix organization and model-curation strategy

The appendix is organized as paired biological and computational tables for each modeling layer. The biological tables document the experimental and mechanistic rationale for including each submodule. The computational tables document candidate mathematical representations, variables, inputs, outputs, and parameterization strategies. Evidence confidence is determined by literature review using the following working categories: established, supported, hypothesized, or model assumption. Table pairs are referenced locally in each module subsection so that biological claims and candidate model forms can be traced back to their evidence and implementation tables.

Pulse Delivery, Exposure Geometry, and Dosimetry

Pulse delivery, exposure geometry, and dosimetry define the engineered exposure that initiates electro-exocytosis. In this layer, the pulse waveform denotes the externally imposed electric-field history, including pulse amplitude, duration, number, polarity, and repetition rate. Early

nsPEF studies show that these variables set the transition from plasma-membrane poration to intracellular Ca^{2+} mobilization, PS exposure, and cell-fate responses^{9,81,99,129}. Exposure geometry denotes the physical arrangement of electrodes, chamber, cuvette, culture format, or flow path that converts a programmed voltage waveform into the electric field experienced by the sample. Because cells may receive different effective exposures depending on electrode gap, edge effects, medium conductivity, cell density, sample volume, and position within the device, geometry is central to reproducibility^{5,42,46}. A nominal pulse protocol is therefore not sufficient unless the exposure configuration, field uniformity, and measurement method are also defined^{52,91}. The biological/process evidence for these exposure variables is summarized in Table A1, and the corresponding computational abstractions are listed in Table A2.

For the present compartmental modeling framework, the spatial field calculation should be treated as an upstream dosimetry and calibration step rather than as a spatially resolved biological state. Device settings and exposure geometry can be collapsed into lumped descriptors such as an effective field amplitude, a cumulative pulse-dose index, a heating constraint or temperature history, an exposure-uniformity factor, and the recovery or harvest interval. These descriptors define controllable engineering inputs and batch-to-batch exposure conditions for the downstream ODE modules, while avoiding the need to track chamber position as a dynamic state variable. This separation is also biologically important: energy deposition and Joule heating must be monitored so that field-driven signaling can be distinguished from thermal artifacts, and the recovery interval must be specified because EV yield, cargo composition, and viability are measured over minutes to hours after pulsing rather than at the instant of exposure. Recent nsPEF spheroid work explicitly links pulse width, repetition rate, buffer conductivity, absorbed energy density, and measured temperature rise, showing that high-conductivity media and compressed pulse trains can produce substantially different thermal histories even when nominal pulse amplitudes are similar⁷⁶. Related cancer-ablation studies emphasize the same practical distinction between absorbed electrical dose and heat retained during the treatment interval³². Figure A1 summarizes this interface between device-level pulse delivery and reduced model inputs.

We therefore use a tiered dosimetry strategy in which increasingly detailed pulse-energy models can be selected according to the available calibration data. The simplest baseline preserves a legacy absorbed-energy descriptor,

$$u_{\text{legacy}} = \sigma E_{\text{peak}}^2 t_p N, \quad (\text{A1})$$

where σ is the medium conductivity, E_{peak} is the nominal peak electric field, t_p is pulse width, and N is pulse number. This expression is useful as a transparent dose index and maintains continuity with early exploratory simulations, but it does not distinguish square, bipolar, or exponentially decaying waveforms and treats geometry correction only as a coarse scalar.

A waveform-aware Joule model retains the same physical skeleton but weights the pulse by the normalized squared

Table 2. Concise module map linking nsPEF-enabled EV engineering biology to appendix model-specification tables.

Layer	Module	Model interface/output	Biology table	Computational table
1	Pulse delivery, exposure geometry, and dosimetry	$E(t)$, pulse width, pulse number, repetition rate, energy, temperature	Table A1	Table A2
2	Plasma membrane and organelle electrodynamics	ΔV_m , ΔV_{ER} , ΔV_{mito} , ΔV_{MVB} , $P_i(t)$	Table A3	Table A4
3	Ion transport, Ca^{2+} mobilization, ROS, and bioenergetics	J_{Ca} , C_i , C_{ER} , C_{mito} , ROS, ATP	Table A5	Table A6
4	Ca^{2+} -dependent remodeling and repair	PS exposure, calpain, annexins, actin state, resealing	Table A7	Table A8
5	EV biogenesis and subtype release	R_{sEV} , $R_{m/lEV}$, R_{AB} , subtype weighting	Table A9	Table A10
6	Cargo sorting, EV composition, and potency	protein/RNA/lipid cargo, antigen enrichment, functional potency	Table A11	Table A12
7	Injury, debris, and quality gate	viability, apoptosis, necrosis, debris fraction, contamination markers	Table A13	Table A14
8	Manufacturing, isolation, and QC interface	yield, recovery, purity, batch consistency, scalability	Table A15	Table A16
Cross-cutting	Cell-state and disease modifiers	parameter scalars, phenotype priors, heterogeneity	Tables A1–A15	Tables A2–A16

Abbreviations: nsPEF, nanosecond pulsed electric field; EV, extracellular vesicle; sEV, small extracellular vesicle; m/lEV, medium/large extracellular vesicle; AB, apoptotic body; PS, phosphatidylserine; ROS, reactive oxygen species; QC, quality control.

Table 3. Modeling framework and module-interface logic for the integrated nsPEF-EV engineering model.

Module	Mathematical representation	Inputs	Outputs	Rationale
Pulse delivery and dosimetry	External forcing, exposure geometry, energy/thermal constraints	pulse protocol, chamber geometry, conductivity, cell density	$E(t)$, dose metrics, temperature	Defines controllable engineering input and batch-to-batch exposure.
Membrane and organelle electrodynamics	Circuit model plus pore-density/pore-radius kinetics	$E(t)$, geometry, membrane properties, organelle size	ΔV_m , ΔV_o , G_m , P_i	Converts nsPEF exposure into plasma-membrane and organelle permeability.
Ion transport, Ca^{2+} , ROS, ATP	Compartmental flux-balance ODEs with optional GHK/Nernst–Planck fluxes	permeability, ion gradients, pumps, channels, stores	C_i , C_{sub} , C_{ER} , C_{mito} , ROS, ATP	Links electroporation to signaling, organelle stress, and recovery capacity.
Remodeling and repair	Reaction–diffusion, buffer kinetics, Hill activation, repair ODEs	Ca^{2+} transient, ROS, membrane damage	effector activation, PS exposure, actin state, resealing	Defines reversible remodeling, budding competence, and membrane recovery.
EV release	Pool-balance and Ca^{2+} -dependent rate models	MVB pool, budding sites, Ca^{2+} , PS, cytoskeletal state	R_{sEV} , $R_{m/lEV}$, R_{AB}	Maps molecular states to size/subtype-resolved vesicle release.
Cargo and potency	Cargo-sorting weights, reaction networks, or phenotype-weighted output models	ESCRT, Rab, tetraspanins, stress, state, EV subtype	protein/RNA/lipid cargo, potency score	Captures EV engineering beyond yield alone.
Injury and quality gate	Threshold rules, hybrid state transitions, contamination models	residual conductance, Ca^{2+} , ATP, ROS, caspases	viable/stressed/apoptotic fractions, debris burden	Separates useful EV release from injury-associated particles.
Manufacturing and QC	Process-yield balances, recovery models, mixed-effects batch models	harvest time, isolation method, QC markers, scale	yield, recovery, purity, reproducibility	Connects cell-level biology to translational EV production.
Cell-state modifiers	Parameter scaling, priors, mixed-effects or ensemble models	cell type, disease state, activation, stiffness, lipid state	module-specific sensitivity factors	Adapts the shared model to different producer cells and disease contexts.

Note: The table defines modeling form at the module level. Appendix tables provide submodule-specific biological evidence and computational implementation options.

Abbreviations: nsPEF, nanosecond pulsed electric field; EV, extracellular vesicle; MVB, multivesicular body; ESCRT, endosomal sorting complex required for transport; PS, phosphatidylserine; ROS, reactive oxygen species; ATP, adenosine triphosphate; QC, quality control; GHK, Goldman-Hodgkin-Katz.

waveform integral,

$$\chi_w = \frac{1}{t_p} \int_0^{t_p} \left(\frac{E(t)}{E_{\text{peak}}} \right)^2 dt, \quad (\text{A2})$$

Prepared using sagej.cls $u_J = \sigma E_{\text{peak}}^2 t_p N \chi_w \langle g_E^2 \rangle, \quad (\text{A3})$

$$\Delta T_{\text{ad}} = \frac{u_J}{C}. \quad (\text{A4})$$

Here χ_w captures waveform shape, $\langle g_E^2 \rangle$ represents the geometry-averaged squared-field factor, and C_v is

the volumetric heat capacity. This adiabatic estimate is intentionally conservative: it represents the temperature rise that would occur if deposited energy remained locally retained. It is appropriate for ranking protocols by electrical energy deposition, but it should not be interpreted as a measured sample temperature when heat diffusion, convection, electrode heat exchange, or long inter-pulse intervals are relevant.

When temperature history is needed, the adiabatic estimate can be coupled to a reduced thermal-retention model,

$$\Delta T_{\text{end}} = \eta \Delta T_{\text{ad}} \frac{\tau_{\text{loss}}}{T_{\text{train}}} \left(1 - e^{-T_{\text{train}}/\tau_{\text{loss}}}\right), \quad (\text{A5})$$

where $T_{\text{train}} \approx N/f_{\text{rep}}$ is the pulse-train duration, τ_{loss} is an empirical thermal relaxation time, and η is a setup-dependent retention factor. This reduced form is not a substitute for finite-element thermal modeling, but it captures the main experimental observation that identical absorbed energy can lead to different end-of-train temperature rises when pulse repetition rate, train duration, buffer conductivity, and exposure geometry differ^{32,76}. Because temperature can itself modulate membrane fluidity, permeabilization thresholds, and cell fate, thermal terms should be reported as independent dosimetry outputs rather than folded silently into an electrical dose scalar^{43,83,112,113,144}.

Plasma Membrane and Organelle Electrodynamics

A central goal of the framework is to connect nsPEF exposure to downstream Ca^{2+} -dependent EV release and cargo modulation without requiring a complete whole-cell reconstruction. The membrane and organelle-electrodynamics layer therefore serves as the upstream bioelectric perturbation module. The applied pulse induces transient plasma-membrane polarization, electropore formation, increased conductance, and ion-specific permeability, all of which have been observed or modeled in single-cell electroporation studies^{45,47,80}. Short pulses can also couple to intracellular membranes, supporting reduced organelle-voltage terms for ER, mitochondrial, endosomal, and MVB-related perturbation^{5,8,99,129}. These outputs provide the primary interface between nsPEF exposure and downstream Ca^{2+} , repair, exocytosis, EV-biogenesis, and cargo-sorting modules.

Computational electroporation literature clarifies how much of this upstream module can remain reduced while still being mechanistically anchored. Molecular dynamics studies show field-driven water defects progressing into hydrophilic pores on nanosecond time scales, with pore formation, water-channel growth, voltage removal, and resealing all occurring as distinct processes^{118,120}. Vesicle electromechanics adds the continuum constraint that breakdown voltage depends on membrane tension, area expansivity, and lipid composition⁷³. Together, these papers support the present choice to keep pore nucleation, pore-radius/tension effects, permeability, and resealing as separable calibrated states rather than one binary permeabilization switch. Table A3 summarizes the biological electrodynamics evidence, and Table A4 lists the reduced computational forms used for this layer.

The current implementation uses a reduced electrodynamics ladder rather than a full spatial pore-kinetic simulation. Plasma-membrane polarization is represented by a Schwan-style limiting voltage scaled by the effective field, cell radius, and a cell-state membrane modifier, with a first-order charging correction for finite pulse width:

$$\begin{aligned} \Delta V_m^\infty &= f_s E_{\text{eff}} a M_m, \\ \Delta V_m(t_p) &= \Delta V_m^\infty (1 - \exp[-t_p/\tau_m]). \end{aligned}$$

where f_s is the Schwan geometry factor, E_{eff} is the geometry-corrected electric field, a is the configurable cell radius, M_m is the membrane-state modifier, t_p is pulse width, and τ_m is an effective membrane charging time^{42,45,46}. This formulation resolves the previous fixed-radius placeholder by making ΔV_m explicitly cell-size dependent; larger cells reach higher induced membrane voltages at the same field, while very short pulses may terminate before the membrane reaches its Schwan limit.

Electroporation is then represented by reduced permeability and pore-density closures:

$$\begin{aligned} P_m &= [1 + \exp[-s(\Delta V_m - V_{\text{th}})]]^{-1}, \\ N_p &= N_{\text{max}} \frac{\Delta V_m^n}{V_{1/2}^n + \Delta V_m^n}. \end{aligned}$$

These terms are phenomenological surrogates for voltage-dependent pore creation, pore growth, conductance, and resealing models^{47,127}. Patch-clamp studies provide direct calibration targets because nsPEF-induced conductance can persist for minutes and recover in a cell-type-dependent manner^{74,80,81}. The reduced pore-density variable should therefore be interpreted as an effective permeability state, not an absolute physical pore count, until it is calibrated against conductance, dye uptake, ion flux, or recovery data. Organelle perturbation is currently represented by configurable fractions of the plasma-membrane voltage, $\Delta V_o = \alpha_o \Delta V_m$ for ER, mitochondria, and MVB/endosomal membranes. This preserves an explicit route from nsPEF exposure to intracellular membrane stress while marking the organelle layer as calibration-dependent until multi-shell, geometry-resolved, or organelle-specific simulations are incorporated^{5,8,99}. The same pore state also feeds the lipid-remodeling layer because experiments and MD simulations connect nanosecond-pulse nanopores with voltage-driven phosphatidylserine translocation^{130,131}.

Ion Transport, Ca^{2+} Mobilization, ROS, and Bioenergetics

The ion-transport and organelle-state module translates nsPEF-induced membrane permeability into intracellular ionic, metabolic, and stress responses. The minimal model treats electropore-mediated Ca^{2+} influx and ER release as source terms in a cytosolic Ca^{2+} balance. More detailed formulations include voltage-gated or leak-channel entry, ER release through IP_3 receptors or ryanodine receptors, SERCA-mediated refilling, PMCA and NCX extrusion, mitochondrial uptake and release, ROS generation, ATP dependence, and osmotic recovery. Table A5 organizes the biological evidence for these submodules, while Table A6

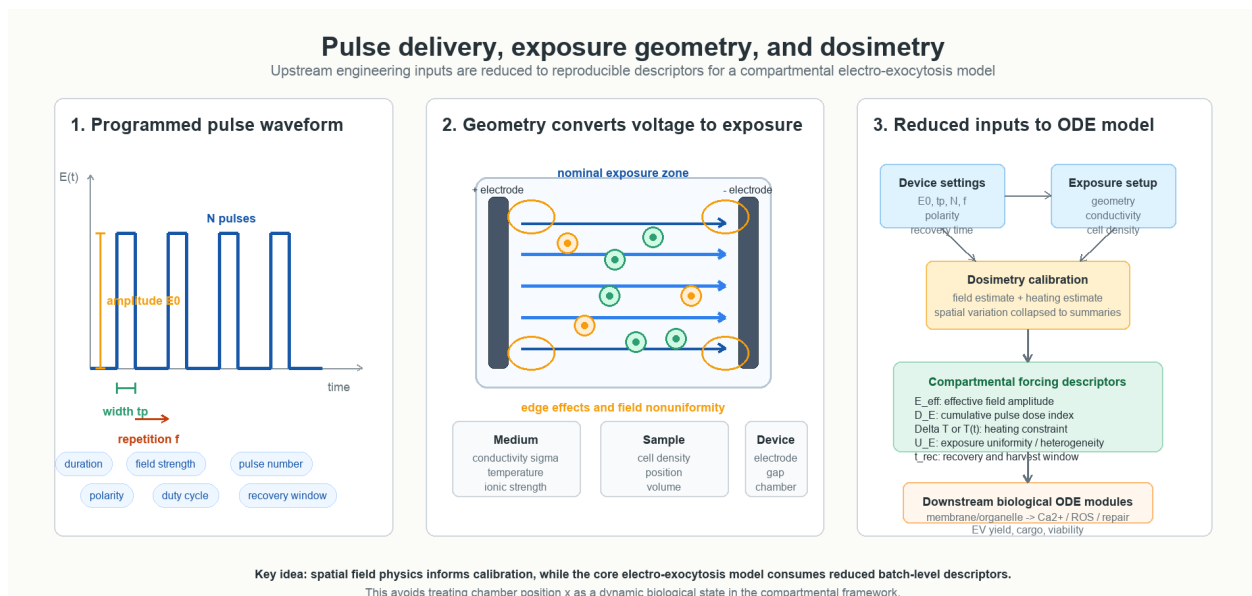


Figure A1. Conceptual interface for pulse delivery, exposure geometry, and dosimetry. The programmed nsPEF waveform and physical exposure configuration determine the actual sample exposure. In a reduced compartmental framework, spatial field information is used for upstream calibration and then summarized as lumped descriptors such as effective field amplitude, cumulative pulse dose, heating constraint, exposure-uniformity factor, and recovery interval. These descriptors provide reproducible inputs to downstream biological ODE modules without requiring a spatially resolved core model.

Abbreviations: nsPEF, nanosecond pulsed electric field; ODE, ordinary differential equation.

Table A1. Biological evidence table for pulse delivery, exposure geometry, and dosimetry.

Submodule	Biological/process phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Pulse waveform	nsPEF duration, amplitude, number, repetition rate	Parameter-response studies across pulse width, field strength, repetition, and pulse number	Defines controllable input space for EV yield and cargo modulation	Established ^{9,76,81,99,129}
Exposure geometry	Electrode/chamber/cuvette geometry	Field uniformity, edge effects, cell density, medium conductivity	Controls reproducibility and batch exposure	Supported; implementation remains lumped ^{5,42,46,52,76}
Energy and heating	Joule heating and thermal artifact risk	Absorbed energy density, waveform energy factor, temperature traces, duty-cycle and repetition-rate effects	Separates bioelectric effects from heat-induced stress	Supported; setup-specific calibration needed ^{32,43,76,112,113}
Recovery interval	Post-pulse harvest window	EV release and repair kinetics over seconds to hours	Determines when yield, cargo, and viability are measured	Supported ^{12,68,104,115,140}
Scale-up mode	Static versus flow-through treatment	Flow exposure, residence time, process monitoring, and sterility constraints	Connects mechanism to manufacturing implementation	Hypothesized ^{6,13,30,53,56}

Abbreviations: nsPEF, nanosecond pulsed electric field; EV, extracellular vesicle.

Table A2. Computational framework table for pulse delivery, exposure geometry, and dosimetry.

Submodule	Model form	Representative relation	Inputs	Outputs
Pulse input	External forcing function ^{9,81,99,129}	$E(t) = E_0 u(t; t_p, N, f)$	amplitude, width, number, repetition	$E(t)$
Dose metric	Selectable absorbed-energy model ^{32,76}	$u = \sigma E_{peak}^2 t_p N \chi_w \langle g_E^2 \rangle$	pulse train, conductivity, waveform, geometry	energy density, dose index
Field distribution	Laplace/quasi-static field model ^{5,42,46,52}	$\nabla \cdot (\sigma \nabla \phi) = 0, E = -\nabla \phi$	chamber geometry, conductivity	local $E(\mathbf{x}, t)$
Thermal constraint	Adiabatic or lumped heat-loss model ^{32,43,76,112,113}	$\Delta T_{end} = F_{ret} \Delta T_{ad}$	field, medium, cooling, pulse train	$T(t), \Delta T_{end}$
Flow-through exposure	Residence-time model ^{13,30,53,56}	$D_E(\mathbf{x}) = \int_{\tau} E(\mathbf{x}(t))^2 dt$	flow rate, chamber field	exposure distribution

Abbreviations: EV, extracellular vesicle.

lists the corresponding reduced flux-balance and organelle-state model forms.

For the current computational framework, this layer is represented as a reduced flux-balance ODE system rather

Table A3. Biological evidence table for plasma membrane and organelle electrodynamics.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Membrane polarization	Rapid transmembrane-potential rise	Measurements or simulations of ΔV_m under nsPEF exposure	Determines whether local membrane regions reach electroporation range	Established; Schwan/charging reduced implementation 5,42,45,46
Plasma-membrane nanopores	Transient nanoscale membrane defects	Conductance, dye uptake, ion permeability, recovery kinetics	Creates pathways for Ca^{2+} and other ion entry	Established; proxy remains phenomenological 46,47,80,81,127
Organelle electroporation	ER, mitochondrial, nuclear, endosomal, and MVB membrane perturbation	Ca^{2+} release, mitochondrial depolarization, intracellular membrane access	Provides route for store release, ROS, and altered MVB/cargo dynamics	Supported; organelle fractions need calibration 5,8,99,129
Permeability recovery	Conductance decay and resealing	Patch clamp, uptake assays, dye exclusion	Sets duration and magnitude of post-pulse ion flux	Established; kinetics are cell-type dependent 74,80,81,83
Isolated EV permeabilization	Direct nsPEF effect on EV membranes	Vesicle integrity, cargo loading, aggregation, marker retention	Defines second mode for post-isolation EV engineering	Hypothesized; not implemented in the current cell-based layer 27,29,37,44,106

Abbreviations: nsPEF, nanosecond pulsed electric field; ER, endoplasmic reticulum; EV, extracellular vesicle; MVB, multivesicular body.

Table A4. Computational framework table for plasma membrane and organelle electrodynamics.

Submodule	Model form	Representative relation	Inputs	Outputs
Membrane charging	Reduced Schwan/circuit model 42,45,46	$\Delta V_m = f_s E_{\text{eff}} a (1 - e^{-t_p/\tau_m})$	$E(t)$, geometry, cell radius, membrane state	ΔV_m
Organelle charging	Reduced multi-shell surrogate 5,8,99	$\Delta V_o = \alpha_o \Delta V_m$	organelle factors, plasma-membrane voltage	ΔV_o
Pore nucleation	Voltage-dependent permeability proxy 118,120	$P_m = [1 + \exp[-s(\Delta V_m - V_{\text{th}})]]^{-1}$	ΔV_m , threshold parameters	P_m
Pore-radius evolution	Pore-energy or reduced kinetic model 73,118	$\frac{dr_p}{dt} = F(r_p, \Delta V, \gamma, \sigma)$	pore energy, tension	$r_p(t)$
Permeability closure	Saturating pore-density closure 47,108	$N_p = N_{\text{max}} \Delta V_m^n / (V_{1/2}^n + \Delta V_m^n)$	pore-density scale, voltage scale, Hill exponent	N_p, P_i
Isolated EV charging	Vesicle-scale circuit model 37,44,106	$C_{EV} \frac{d\Delta V_{EV}}{dt} + G_{EV} \Delta V_{EV} = f(E, r_{EV})$	EV radius, medium conductivity	$P_{EV}(t)$, integrity state

Abbreviations: ER, endoplasmic reticulum; EV, extracellular vesicle; MVB, multivesicular body.

than a full electrodiffusion model. The pore term is treated as a transient extracellular Ca^{2+} source gated by the electrodynamic permeability state. This choice is supported by studies in which nsPEF rapidly elevates cytosolic Ca^{2+} and the response depends on extracellular Ca^{2+} , pulse width, and membrane pathway [4,129,146](#). In CHO cells, a 60-ns pulse produced separable external- Ca^{2+} and ER-release thresholds and a critical Ca^{2+} range for amplification, making these data especially useful for tuning reduced Ca^{2+} source terms [99](#).

ER mobilization is modeled as an electrically triggered store-release pathway with SERCA refilling, while IP₃R/RyR-mediated amplification is retained as a cell-type-specific extension for excitable or store-sensitive cells [3,100,123,124,135](#). The computational calcium literature gives useful bounds for that extension: Li and Rinzel reduced a detailed IP₃ receptor model to a Hodgkin–Huxley-like form with IP₃ regulation, Ca^{2+} activation, and Ca^{2+} inactivation, while Shannon and coworkers used junctional, subsarcolemmal, and bulk compartments with realistic buffering, reversible SR uptake, Na^+ – Ca^{2+} exchange, and load-dependent release [54,101](#). We therefore keep the default ER term compact, but expose channel opening, store content, pump capacity, and exchanger recovery as parameters when cell-type-specific data are available.

Na^+ , K^+ , and Cl^- fluxes are included as lumped electropore/channel-mediated perturbations that relax toward baseline and drive an osmotic-stress state. This abstraction is justified by evidence that nsPEF increases leak conductance, depolarizes membrane potential, alters voltage-gated channels, and changes cell volume [17,81,109,145](#). Na^+/K^+ -ATPase sensitivity is retained as a recovery modifier because ATP1A1 silencing attenuates membrane disruption after nanosecond pulses [105](#).

Mitochondrial Ca^{2+} uptake/release and depolarization connect cytosolic Ca^{2+} to ROS and ATP availability. nsPEF studies support mitochondrial membrane perturbation and ROS generation, while broader calcium-electroporation data show that Ca^{2+} overload can deplete ATP and shift the cell toward injury [8,31,71,75,84,89](#). On the computational side, mitochondrial modules can be scaled from a minimal Ca^{2+} -handling block with respiration, ATP synthase, proton leak, adenine nucleotide exchange, uniporter, and Na^+ – Ca^{2+} exchange [59](#), to a simplified ATP-production model driven by glycolytic and Ca^{2+} inputs [11](#), or to a thermodynamically balanced oxidative-phosphorylation model that conserves mass, charge, and free energy [7](#). ROS is represented with Ca^{2+} -, mitochondrial-, osmotic-, and pulse-dependent source terms plus first-order clearance; this reduced form is consistent with models in which ROS

production, scavenging, and ROS-sensitive inner-membrane conductance shape mitochondrial potential, NADH, and redox dynamics^{23,70,148}. ATP is kept as a normalized production-consumption state coupled to mitochondrial potential, pump activity, repair demand, and oxidative stress.

$$\frac{d[\text{Ca}^{2+}]_i}{dt} = J_{\text{Ca,pore}} + J_{\text{VGCC}} + J_{\text{ER,rel}} - J_{\text{SERCA}} - J_{\text{PMCA}} - J_{\text{NCX}} - J_{\text{mito}} - J_{\text{buffer}}. \quad (\text{A6})$$

Ca²⁺-Dependent Remodeling and Repair

This module connects global and submembrane Ca²⁺ signals to local membrane remodeling and repair. Although cytosolic Ca²⁺ may be represented as a well-mixed compartment in reduced models, electro-exocytosis is likely governed by local Ca²⁺ domains near pores, damaged membrane regions, budding sites, repair complexes, and fusion machinery. Reaction-diffusion theory provides a practical reason to distinguish these domains from the bulk pool: mobile and immobile buffers can confine Ca²⁺ elevations over nanometer-to-submicrometer distances, with local gradients forming on submillisecond time scales near an open channel⁷². Ca²⁺-dependent resealing occurs on experimentally measurable time scales, with nano- and microsecond electroporation studies reporting fast and slow repair components that are sensitive to extracellular Ca²⁺¹². Local Ca²⁺ therefore serves not only as an influx variable but also as the organizing signal for lipid asymmetry, cytoskeletal mechanics, curvature, scission, fusion competence, and resealing. The paired evidence and modeling assumptions for these repair and remodeling variables are tabulated in Tables A7 and A8.

For the current computational framework, this layer is implemented as a reduced Ca²⁺-sensor and repair-state module. The upstream ion-transport state is converted into a submembrane Ca²⁺ microdomain proxy amplified by pore activation and osmotic stress, with buffer strength and shell depth treated as tunable parameters rather than fixed geometry. This choice reflects evidence that ESCRT and repair machinery assemble at damaged membrane domains rather than responding only to bulk cytosolic Ca²⁺^{96,102}. Annexin recruitment is represented as a Ca²⁺-dependent membrane-binding and curvature-stabilization state because annexin A4/A6/A7 can support membrane constriction, ESCRT-associated repair, and repair-linked vesicle secretion^{15,69,111,140}. Molecular and biophysical annexin studies then provide the curvature and lipid-composition rationale for keeping annexin effects separate from generic repair activity^{34,67}.

The lipid-remodeling branch maps local Ca²⁺ to scramblase activation, flippase suppression, and PS externalization. This branch is treated as established because nsPEF activates TMEM16F-dependent PS exposure and blebbing, while newer TIRF studies show delayed, spatially punctate PS externalization after 400-ns pulses^{68,104}. The Vernier studies add a second, faster computationally tractable route: nanopore-facilitated, voltage-driven PS movement can occur along electropore walls and appears experimentally as

localized, anode-facing PS externalization after nanosecond pulses^{130,131}. The model therefore keeps an immediate pore/voltage PS term separate from slower Ca²⁺-scramblase dynamics. Chemically controlled TMEM16F vesiculation, lipid-asymmetry loss, and Ca²⁺-regulated flippases provide complementary support for linking Ca²⁺ and PS state to EV-like budding competence^{35,98,134}.

The cytoskeletal branch maps local Ca²⁺ to calpain activity, actin disruption, actomyosin tension, and repair-associated shedding. Recent nsPEF actin and swelling studies are especially useful here because Ca²⁺-dependent actin shrinkage and swelling can be tested directly against the model's actin-disruption and osmotic-stress terms^{36,82,90}. Calpain and ezrin-cleavage studies support the protease component, and membrane-repair studies show that calpain can orchestrate annexin-containing microvesicle release^{18,92,141}.

Finally, lysosomal repair is represented as a Ca²⁺-triggered vesicle-fusion and membrane-resealing support state. Lysosomal exocytosis, extracellular protease release, sphingolipid-dependent EV biology, lysosomal-channel regulation, and ESCRT-mediated lysosome repair support this reduced repair branch^{2,21,88,97,128}. The resulting variables—PS exposure, calpain activity, annexin activity, lysosomal repair, actin state, resealing state, and repair-associated shedding—serve as interpretable coupling terms for medium/large EV budding, membrane recovery, and injury-quality gating.

EV Biogenesis and Subtype Release

The EV-release module converts nsPEF-induced intracellular and membrane responses into measurable vesicle outputs. It integrates endosomal/MVB-derived small-EV release, plasma-membrane budding of medium/large EVs, and apoptotic body or injury-associated vesiculation. The model should retain mechanistic terms such as exosome and microvesicle internally, but report outputs using EV nomenclature compatible with the level of experimental characterization. The biological release mechanisms are organized in Table A9, with candidate pool-balance and rate-law implementations listed in Table A10.

Small-EV production begins with endosome maturation and ILV biogenesis. Recent Rab-conversion studies indicate that Rab5-to-Rab7 transition is not an isolated timing event, but is coordinated with ESCRT-mediated cargo corraling through the Mon1/Ccz1 axis, so maturation and secretory competence should be treated as coupled states rather than as independent switches^{14,77,110}. Within maturing endosomes, canonical ESCRT machinery supplies one major ILV route: ESCRT-0 captures ubiquitinated cargo through Hrs/STAM, ESCRT-I/II concentrate cargo and deform the bud neck, and ESCRT-III/Vps4 completes constriction and scission^{87,116,142}. However, exosome-biased ILV formation is not exclusively canonical. ALIX-ESCRT-III, ceramide, and tetraspanin-rich membrane domains provide selective alternative routes that can redirect cargo and alter whether MVBs are routed toward secretion or lysosomal degradation^{22,25,50,93,132}.

Release of those ILVs as extracellular small EVs then depends on trafficking, docking, and fusion at the cell cortex.

Table A5. Biological evidence table for ion transport, Ca^{2+} mobilization, ROS, and bioenergetics.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Electropore influx	Ca^{2+} Rapid extracellular entry	Ca^{2+} imaging, chelation, Ca^{2+} -free media	Primary trigger for downstream release and repair mechanisms	Established ^{4,12,19,66,146}
ER Ca^{2+} release	Intracellular store mobilization	$\text{IP}_3\text{R}/\text{RyR}$ involvement, ER store depletion	Amplifies cytosolic Ca^{2+} transients and fusion signals	Established ^{3,100,123,124,135}
Na^+ , K^+ , Cl^- fluxes	Ionic/osmotic perturbation	Membrane-potential recovery, swelling, ionic imbalance	Modulates volume, tension, and repair state	Supported ^{17,68,105,109,145}
Mitochondrial Ca^{2+}	Uptake/release and depolarization	Mitochondrial Ca^{2+} , $\Delta\Psi_m$, mPTP evidence	Links Ca^{2+} to metabolic stress and fate decisions	Supported ^{8,70,71}
ROS generation	Oxidative stress after nsPEF	ROS probes, antioxidant rescue, mitochondrial source evidence	Modifies EV secretion, cargo, and injury transition	Supported ^{51,75,84,139,148}
ATP/bioenergetics	Energy availability for pumps and trafficking	ATP depletion, pump recovery, metabolic assays	Controls repair, cargo sorting, and EV-release capacity	Hypothesized ^{28,31,89}

Abbreviations: EV, extracellular vesicle; ER, endoplasmic reticulum; IP_3R , inositol 1,4,5-trisphosphate receptor; RyR, ryanodine receptor; ROS, reactive oxygen species; ATP, adenosine triphosphate; mPTP, mitochondrial permeability transition pore.

Table A6. Computational framework table for ion transport, Ca^{2+} mobilization, ROS, and bioenergetics.

Submodule	Model form	Representative relation	Inputs	Outputs
Pore-mediated Ca^{2+} flux	GHK/Ohmic flux ^{118,120}	$J_{\text{Ca,pore}} = P_{\text{Ca}}(t)\Phi_{\text{Ca}}(C_e, C_i, V_m)$	P_{Ca} , gradients, voltage	Ca^{2+} source term
ER release	Store-release ODE ^{54,101}	$J_{\text{ER,rel}} = k_{\text{rel}}O_{\text{IP3R/RyR}}(C_i, C_{\text{ER}})$	channel state, store content	ER-to-cytosol flux
Pump/exchanger recovery	Saturating fluxes ¹⁰¹	$J_{\text{PMCA}} = V_{\text{max}} \frac{C_i^n}{K^n + C_i^n}$	Ca^{2+} , Na^+ , ATP, voltage	Ca^{2+} clearance
Mitochondrial buffering	Coupled uptake/release ODE ^{11,59}	$\frac{dC_m}{dt} = J_{\text{MCU}} - J_{\text{NCLX}} - J_{\text{mPTP}}$	C_i , $\Delta\Psi_m$, ROS	mitochondrial Ca^{2+} state
ROS dynamics	Source-clearance ODE ²³	$\frac{dR}{dt} = k_{\text{ROS}}f(C_m, \Delta\Psi_m) - k_{\text{clear}}R$	mitochondrial state, antioxidants	ROS state variable
ATP state	Production-consumption ODE ^{7,11}	$\frac{dA}{dt} = k_{\text{prod}}f(\Delta\Psi_m) - k_{\text{use}}(J_{\text{pump}} + R_{\text{repair}})$	mitochondria, pumps, repair	ATP availability

Abbreviations: ER, endoplasmic reticulum; GHK, Goldman-Hodgkin-Katz; PMCA, plasma membrane Ca^{2+} ATPase; MCU, mitochondrial calcium uniporter; NCLX, $\text{Na}^+/\text{Ca}^{2+}/\text{Li}^+$ exchanger; ROS, reactive oxygen species; ATP, adenosine triphosphate; mPTP, mitochondrial permeability transition pore.

Table A7. Biological evidence table for Ca^{2+} -dependent remodeling and repair.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Ca^{2+} microdomains	Local Ca^{2+} elevation near pores and membranes	High-speed/local Ca^{2+} imaging, buffer sensitivity	Localizes activation to budding, repair, and fusion sites	Supported ^{96,102}
Annexin recruitment	Ca^{2+} -dependent membrane binding	Annexin translocation and repair studies	Links Ca^{2+} to membrane repair and curvature stabilization	Supported ^{15,34,67,69,111}
Scramblase/PS exposure	Loss of lipid asymmetry	Annexin V staining, scramblase/flippase studies	Promotes MV-like budding and EV surface phenotype	Established ^{35,68,98,104,134}
Calpain activation	Ca^{2+} -dependent protease activity	Calpain activation/inhibition, cytoskeletal remodeling	Supports blebbing, scission, and membrane remodeling	Supported ^{18,92,141}
Actomyosin remodeling	Tension, contraction, and scission	RhoA/ROCK, myosin, actin perturbation	Determines budding probability and vesicle detachment	Supported ^{36,82,90}
Lysosomal repair	Ca^{2+} -triggered vesicle fusion	LAMP1 exposure, lysosome fusion	Adds membrane and may release vesicle-like material	Supported ^{2,21,88,97,128}

Abbreviations: EV, extracellular vesicle; PS, phosphatidylserine; ESCRT, endosomal sorting complex required for transport.

Rab27, Rab11, and Rab35 families regulate MVB positioning and membrane access, while Munc13-4 provides a Ca^{2+} -responsive trafficking and priming branch that helps generate release-competent MVBs ^{38,40,61,65,117}. The terminal fusion

Table A8. Computational framework table for Ca^{2+} -dependent remodeling and repair.

Submodule	Model form	Representative relation	Inputs	Outputs
Ca^{2+} microdomain	Reaction–diffusion shell ODE ^{72,101}	$\frac{\partial C}{\partial t} = D_{Ca} \nabla^2 C + J_{Ca} - R_{buffer} - R_{effector}$	local flux, diffusion, buffers	$C_{sub}(t)$
Generic Ca^{2+} sensor	Hill activation ^{72,96,101}	$A_x = \frac{C_{sub}^n}{K_x^n + C_{sub}^n}$	local Ca^{2+} , sensitivity	activated fraction
PS exposure	Activation plus lipid-state ODE ^{68,130,131}	$\frac{dPS_{out}}{dt} = k_{scr} A_{scr} (1 - PS_{out}) - k_{flip} PS_{out}$	Ca^{2+} , scramblase, flip-pase	PS externalization
Calpain/cytoskeleton	Protease activation and substrate loss ^{18,92,141}	$\frac{dF_{actin}}{dt} = -k_{calp} A_{calp} F_{actin} + k_{poly}$	Ca^{2+} , calpain state	cortical remodeling
Membrane resealing	Conductance-recovery ODE ^{73,118}	$\frac{dG_m}{dt} = -k_{seal} A_{repair} G_m$	residual conductance, repair activity	resealed membrane state
Damaged-domain shedding	Damage-removal rate ⁷³	$R_{shed} = k_{shed} D_m f_{Ca} f_{tension}$	damaged membrane, Ca^{2+} , tension	repair-associated shedding

Abbreviations: PS, phosphatidylserine; ODE, ordinary differential equation.

step is mediated by cell-context-dependent SNARE assemblies, with syntaxin-4/SNAP-23/VAMP-7 and VAMP5-based complexes now supported by direct exosome-secretion studies, and with synaptotagmin-7-like Ca^{2+} sensing providing a plausible fusion-pore stabilization mechanism under strong submembrane Ca^{2+} signals ^{24,57,62,143}. In parallel, outward budding of medium/large EVs is supported by ARF-, RhoA/ROCK-, phosphatidylserine-, and actomyosin-dependent membrane remodeling, whereas apoptotic bodies arise from caspase-linked ROCK1 activation and repetitive bleb fragmentation during higher-injury states ^{26,63,78,85,94,121}. These mechanistic branches justify treating subtype release as a competition between secretory MVB fate, plasma-membrane budding competence, and apoptotic disassembly rather than as one generic EV-rate variable.

For the current computational framework, this layer is implemented as a reduced pool-balance model with explicit MVB, ILV, docked-MVB, budding-precursor, and apoptotic-commitment states. Endosome maturation is represented as a Rab-conversion-supported source term for a secretory MVB pool, ILV loading is split into ESCRT-dependent and ceramide/tetraspanin-assisted branches, and MVB fate is biased between secretion and lysosomal loss by an acidification-sensitive routing term ^{22,77,132}. Small-EV release is then written as a Ca^{2+} - and SNARE-sensitive fusion flux from the docked MVB pool, while medium/large EV release is represented as a phosphatidylserine-, calpain-, and actomyosin-weighted budding/scission process. Apoptotic-body release is modeled separately as a damage- and blebbing-weighted branch so that injury-associated vesiculation is not conflated with productive secretory EV release ^{57,65,85,121}. This reduced formulation matches the current code implementation while preserving a direct path for later replacement with cell-type-specific Rab, SNARE, lipid, or apoptosis submodels as quantitative data become available.

State-Conditioned EV Size–Composition and Observation Bridge

The release states above specify biological routes, whereas particle-sizing assays report operational diameter classes. These descriptions should remain distinct because endosomal, plasma-membrane, and apoptotic pathways can generate overlapping size ranges, and diameter alone does not establish vesicle origin ¹³⁸. We therefore introduce an

inter-layer bridge rather than redefining the existing sEV, m/LEV, and AB states. A biological mode q denotes ESCRT-enriched MVB/ILV release, ceramide/tetraspanin-enriched MVB/ILV release, plasma-membrane shedding or repair-associated release, or apoptotic fragmentation. The first two modes sum to the existing R_{sEV} output, the plasma-membrane mode contributes to $R_{m/LEV}$, and the apoptotic mode contributes to R_{AB} . This separation is consistent with evidence that activation can differentially regulate distinct nanosized populations released by CD4^+ T cells ¹²⁶.

Let $\mathbf{x}(t)$ denote the upstream cell-state vector, d vesicle diameter, and $\mathbf{c}$ the composition vector. The joint product kernel for biological mode q is factorized as

$$K_q(d, \mathbf{c} | \mathbf{x}) = f_q(d | \boldsymbol{\theta}_q[\mathbf{x}]) g_q(\mathbf{c} | d, \mathbf{x}),$$

where f_q is a nonnegative birth-size distribution and g_q represents conditional cargo allocation. Positive right-skewed distributions are appropriate starting points. For MVB-derived small EVs, a dynamic-scaling distribution derived from membrane-domain growth and ILV budding provides a directly implementable candidate; its tail exponent was sensitive to signaling perturbation ⁸⁶. More detailed ESCRT/cargo-crowding membrane models can constrain plausible ILV sizes and coupling signs but introduce unobserved mechanical parameters that are unlikely to be identifiable from sparse longitudinal particle counts ⁵⁵. Broad lognormal or gamma distributions are therefore retained as reduced alternatives for plasma-membrane and apoptotic modes. Their baseline median and geometric standard deviation should be cell- and assay-specific priors rather than universal subtype constants.

State dependence can be introduced parsimoniously through distribution parameters,

$$\log d_{50,q}(t) = \alpha_q + \beta_q^T [\mathbf{z}_q(t) - \mathbf{z}_{q,0}],$$

where $\mathbf{z}_q$ contains only pathway-matched states. Candidate inputs include ESCRT and ceramide/secretory routing for MVB modes; PS exposure, calpain, actomyosin, membrane tension, and repair shedding for plasma-membrane release; and damage, apoptosis, and blebbing for the apoptotic mode. The nested null case $\beta_q = \mathbf{0}$ changes pathway mixture weights while retaining fixed within-pathway size distributions.

Table A9. Biological evidence table for EV biogenesis and subtype release.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Endosome/MVB maturation	Early-to-late endosome and MVB formation	Rab5/Rab7, endosome maturation, ILV evidence	Defines small-EV-producing compartment pool	Established ^{14,77,110}
ESCRT-dependent ILV formation	Cargo sorting and ILV biogenesis	ESCRT, ALIX, TSG101 perturbation	Generates exosome-enriched precursors	Established ^{50,87,116,142}
ESCRT-independent ILV formation	Ceramide/tetraspanin/lipid-domain sorting	nSmase2, ceramide, CD63/CD81/CD9 evidence	Alternative small-EV biogenesis route	Supported ^{22,25,93}
MVB trafficking/docking	Rab-dependent transport to plasma membrane	Rab27A/B, Rab11, Rab35 evidence	Controls release-ready MVB pool	Established ^{38,40,61,65,117}
MVB fusion	SNARE-mediated fusion with plasma membrane	SNARE and Ca ²⁺ -sensitive fusion evidence	Converts ILVs into secreted small EVs	Supported ^{24,57,62,143}
Plasma membrane budding	MV-like outward budding	ARF6/RhoA/PS/cytoskeleton studies	Produces medium/large EV precursors	Supported ^{26,63,85}
Apoptotic body formation	Blebbing and fragmentation	Caspase/ROCK1/apoptosis evidence	Captures high-dose or injury-associated vesicles	Established ^{78,94,121}

Abbreviations: EV, extracellular vesicle; MVB, multivesicular body; ILV, intraluminal vesicle; ESCRT, endosomal sorting complex required for transport.

Table A10. Computational framework table for EV biogenesis and subtype release.

Submodule	Model form	Representative relation	Inputs	Outputs
MVB pool	Compartmental pool balance ^{14,39,77,110}	$\frac{dM}{dt} = k_{mat}E_{endo} - k_{dock}M - k_{deg}M$	endosome pool, maturation	MVB number
ILV loading	Production-rate law ^{22,25,50,87,116,142}	$\frac{dI}{dt} = k_{ILV}f_{ESCRT}M - k_{release}I$	ESCRT/ceramide state, MVB pool	ILV load
MVB docking	Trafficking/docking ODE ^{38,40,61,65,117}	$\frac{dM_d}{dt} = k_{dock}f_{Rab}M - k_{fus}M_d$	Rab activity, actin barrier	docked MVB pool
Small-EV release	Ca ²⁺ -sensitive fusion rate ^{24,57,62,143}	$R_{sEV} = k_{fus}f_{Ca}(C_{sub})M_d$	docked MVBs, Ca ²⁺ , SNARE	small-EV release rate
Budding-site formation	Budding ODE ^{26,63,85}	$\frac{dB_{m/IEV}}{dt} = k_{bud}f_{PS}f_{tension} - k_{scission}B_{m/IEV}$	PS, curvature, tension	budding precursor pool
Medium/large EV release	Scission-rate law ^{26,63,85,141}	$R_{m/IEV} = k_{scission}f_{actomyosin}B_{m/IEV}$	calpain, ROCK, actomyosin	medium/large EV rate
Apoptotic vesicle release	Injury-state weighted rate ^{20,78,94,121}	$R_{AB} = k_{AB}F_{apop}$	apoptotic fraction, blebbing rate	apoptotic-body release

Abbreviations: MVB, multivesicular body; ILV, intraluminal vesicle; ESCRT, endosomal sorting complex required for transport; sEV, small extracellular vesicle; m/IEV, medium/large extracellular vesicle; AB, apoptotic body.

For true-diameter bin $b = [d_b^-, d_b^+]$, the per-cell birth source and extracellular stock are

$$S_{qb}(t) = R_q(t) \int_{d_b^-}^{d_b^+} f_q(d | \theta_q[\mathbf{x}(t)]) dd,$$

$$\frac{dN_{qb}}{dt} = N_{cells}S_{qb}(t) - k_{loss,q}(d_b)N_{qb} + \sum_{b'} T_{bb'}N_{qb'}.$$

This sectional population balance can use many numerical bins while retaining few shared parameters. It also preserves cohort history: particles remain at their emitted diameter unless an explicit aggregation or fragmentation term $T_{bb'}$ transfers them. Initially, a common effective loss rate and $T_{bb'} = 0$ provide the most identifiable model.

Cargo is linked jointly to pathway, state, and size rather than determined by diameter alone. A reduced geometric capacity relation is

$$E[C_j | d, q, \mathbf{x}] = a_{jq}(\mathbf{x})d^2 + b_{jq}(\mathbf{x})d^3,$$

where the d^2 and d^3 terms represent membrane-associated and luminal capacity, respectively ¹⁴⁷. The coefficients retain pathway- and state-dependent sorting, and low-copy cargo can be represented by stochastic occupancy.

Single-vesicle measurements showing selective changes in EV size, membrane order, and cargo after Rab27a or CD63 perturbation support this conditional rather than deterministic representation ¹³³.

Finally, biological stocks are mapped to an observed size bin b in assay run r through

$$\hat{y}_{rb}(t) = B_{rb} + \sum_{\ell} H_{rb\ell} \eta_{r\ell} \frac{\sum_q N_{q\ell}(t)}{V_r},$$

where $H_{rb\ell}$ represents true-to-measured diameter error and rebinning, $\eta_{r\ell}$ represents size-dependent recovery, B_{rb} is background, and V_r is sample volume. This observation operator is required because different single-particle methods can report materially different size distributions and concentrations for the same EV preparation ^{125,138}. Instrument-response parameters should be constrained by calibration particles, instrument settings, and orthogonal characterization rather than inferred freely with the biological kernel.

For sparse longitudinal data, calibration should begin with literature- or sham-informed fixed kernels, dynamic pathway mixture weights, and a common extracellular loss term. A state-to-size coefficient should be retained only if it improves

prediction for held-out pulse conditions; size-dependent loss should then be tested separately. Particle counts alone cannot identify pathway-specific cargo coefficients, which require marker-resolved or molecular measurements.

Cargo Sorting, EV Composition, and Potency

The cargo module represents the major conceptual expansion from EV-release modeling to EV-engineering modeling. nsPEF may alter EV composition through Ca^{2+} -dependent trafficking, stress-responsive transcription and translation, lipid remodeling, ESCRT/tetraspanin/ceramide pathways, RNA-binding proteins, and injury-state contamination. Therefore, EV output should be represented as a multi-dimensional product state rather than as total particle number alone. Table A11 separates the biological cargo-sorting evidence, and Table A12 specifies the computational cargo, composition, potency, and direct-loading model options.

The implemented Layer 6 module now follows the table structure explicitly. It computes a subtype-weighted cargo state from small-EV, medium/large-EV, and apoptotic-body outputs; separates protein, RNA, lipid, antigen, and direct-loading terms; and maps the resulting cargo vector to a saturating potency score. This keeps the implementation aligned with evidence that EV subtypes carry distinct protein compositions^{41,48}, that RNA sorting can be mediated by RNA-binding proteins such as YBX1 and SYNCRIP^{58,95,103}, that lipid composition and ceramide biology shape EV state¹⁰⁷, and that direct EV loading can be represented separately from endogenous cargo sorting^{27,29}.

Injury, Debris, and Quality Gate

The injury-quality module defines whether nsPEF exposure produces reversible EV engineering, stress-associated release, apoptosis, necrosis, or contaminating debris. This layer is essential because increased particle number is not necessarily equivalent to increased useful EV production. The quality gate separates bona fide EV output from apoptotic bodies, damaged organellar fragments, protein aggregates, and other co-isolated materials. Table A13 summarizes the biological and QC evidence for this distinction, while Table A14 defines the damage, fate-transition, mixture, and constrained-optimization model forms.

The implemented Layer 7 module expands the previous binary quality proxy into a mixture-aware gate. Damage is mapped to stressed-viable, apoptotic, and necrotic fractions using threshold-like Hill terms inspired by systems models of cell-death switching^{1,10}. The measured particle pool is then decomposed into bona fide EVs, apoptotic bodies, debris, and aggregate-like contaminants, which makes the purity score sensitive to both injury state and non-EV co-isolates. This structure matches EV-composition and isolation cautions from apoptotic/necroptotic EV studies and MISEV-style purity guidance^{16,20,137,138}.

Manufacturing, Isolation, and Quality-Control Interface

The manufacturing module links cell-level mechanisms to translational EV production. It represents harvest timing, cell number, culture format, isolation recovery, marker-based QC, purity, batch variability, and flow-through scalability.

This module is intentionally placed after biological release and quality gating because manufacturing success depends on both biological production and downstream process recovery. Table A15 summarizes the process and QC evidence to curate, and Table A16 lists the process-yield, recovery, batch, and optimization model forms.

The implemented Layer 8 module converts biological output into process-facing quantities: cell-normalized yield, harvest-rate-equivalent accumulation, isolation recovery, process purity, batch-adjusted yield, and a constrained optimization objective. This keeps the code aligned with EV manufacturing literature emphasizing scalable source material, recovery and purity tradeoffs, clinical-grade purification, and validation requirements^{64,79,119,136,138}. The objective is intentionally transparent rather than prescriptive: potency, yield, purity, and viability weights can be changed for different process-development priorities.

Cell-State and Disease-State Modifiers as Cross-Cutting Parameters

Cell-state and disease-state modifiers alter how a given cell responds to the same nsPEF exposure. In this framework, they do not define a separate downstream pathway. Instead, they scale or reshape the behavior of all modules. Relevant modifiers include membrane composition, cholesterol and sphingolipid content, cortical stiffness, cytoskeletal prestress, baseline Ca^{2+} handling, mitochondrial state, metabolic capacity, oxidative-stress tone, inflammatory signaling, hypoxia response, baseline EV-biogenesis activity, and cargo-sorting phenotype. Because these modifiers act across layers, their evidence and parameter-scaling implications are distributed across the paired module tables summarized in Table 2 and specified in Tables A1–A16.

$$\theta^{cell} = S_{cell}\theta^{base}, \quad (\text{A7})$$

$$\theta_i \sim \mathcal{D}(\mu_{cell}, \Sigma_{cell}), \quad (\text{A8})$$

where θ^{cell} represents cell-specific parameter values, S_{cell} represents phenotype-specific scaling factors, and $\mathcal{D}(\mu_{cell}, \Sigma_{cell})$ represents population heterogeneity.

Core Governing Equations for a First Integrated Implementation

A first reduced implementation can be constructed by coupling pulse exposure, membrane permeability, Ca^{2+} balance, remodeling activation, EV-release rates, cargo state, repair, and quality gating. The following equations are placeholders for model assembly and should be refined as literature-supported flux expressions and parameters are identified. They assemble the computational forms listed in Tables A2, A4, A6, A8, A10, A12, and A14; manufacturing-facing optimization is summarized separately in Table A16.

$$C_m \frac{d\Delta V_m}{dt} + G_m(t)\Delta V_m = f(E(t), \theta), \quad (\text{A9})$$

$$C_o \frac{d\Delta V_o}{dt} + G_o(t)\Delta V_o = f_o(E(t), r_o, \theta_o), \quad (\text{A10})$$

Table A11. Biological evidence table for cargo sorting, EV composition, and potency.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Protein sorting	Ubiquitin/ESCRT/tetraspanin-dependent protein enrichment	ALIX, TSG101, CD63/CD81/CD9, antigen enrichment	Determines therapeutic or immunogenic protein cargo	Established ^{41,48,50,93,116}
RNA sorting	RNA-binding protein-mediated miRNA/mRNA loading	hnRNPA2B1, SYNCRIP, YBX1, motif-based sorting	Enables nucleic-acid cargo modulation	Supported ^{58,95,103}
Lipid sorting	Ceramide, cholesterol, sphingolipid, PE/PS enrichment	Lipidomics, nSMase2, tetraspanin domains	Controls EV stability, curvature, uptake, immunologic function	Established ^{22,25,93,107}
Stress-responsive cargo	ROS, hypoxia, inflammatory, apoptotic cargo changes	proteomics/lipidomics/RNA-seq after nsPEF	Distinguishes programmed cargo modulation from debris	Supported ^{20,68,84}
Antigen enrichment	Tumor antigen expression and EV loading	MAGE and other antigen studies	Supports vaccine-oriented EV engineering	Supported ^{49,114}
Functional potency	Recipient-cell response	uptake, cargo delivery, immune activation, repair assays	Connects cargo state to biological activity	Hypothesized ^{29,33,122}

Abbreviations: EV, extracellular vesicle; ESCRT, endosomal sorting complex required for transport; ALIX, ALG-2-interacting protein X; TSG101, tumor susceptibility gene 101; miRNA, microRNA; mRNA, messenger RNA; PE, phosphatidylethanolamine; PS, phosphatidylserine; RNA-seq, RNA sequencing.

Table A12. Computational framework table for cargo sorting, EV composition, and potency.

Submodule	Model form	Representative relation	Inputs	Outputs
Subtype-weighted cargo	Weighted model ^{20,41,48}	output $Cargo_j = w_{j,sEV} R_{sEV} + w_{j,m/IEV} R_{m/IEV} + w_{j,AB} R_{AB}$	subtype rates, cargo weights	cargo abundance
Stress-modulated sorting	Rate law ^{58,84,95,103}	$\frac{dCargo_j}{dt} = k_{sort,j} f_{stress} f_{Ca} f_{ESCRT} - k_{loss,j} Cargo_j$	Ca ²⁺ , ROS, ESCRT, stress	cargo state
Antigen enrichment	Transcription/sorting coupling ^{49,114}	$\frac{dA_g}{dt} = k_{expr} f_{stress} - k_{deg} A_g - k_{sort} A_g$	stress, antigen expression	antigen cargo
Lipid composition	Composition-state ODE ^{22,25,107}	$\frac{dL_j}{dt} = k_{syn,j} f_{stress} - k_{sort,j} L_j$	lipid metabolism, EV subtype	lipid cargo class
Potency mapping	Empirical or mechanistic response model ^{29,33,122}	$Potency = F(Cargo, Subtype, Dose)$	cargo vector, recipient assay	functional potency
Direct EV loading	Permeability-diffusion model ^{27,29,37,44}	$\frac{dC_{load}}{dt} = P_{EV}(t) A_{EV} (C_{ext} - C_{EV}) - k_{leak} C_{EV}$	EV permeability, cargo gradient	loaded cargo

Abbreviations: EV, extracellular vesicle; ESCRT, endosomal sorting complex required for transport; ALIX, ALG-2-interacting protein X; sEV, small extracellular vesicle; m/IEV, medium/large extracellular vesicle; AB, apoptotic body.

Table A13. Biological evidence table for injury, debris, and quality gating.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Viability preservation	Reversible response versus cell death	viability, membrane recovery, clonogenic/metabolic assays	Defines acceptable EV-production window	Established ^{8,12,83}
Mitochondrial injury	Depolarization, mPTP, cytochrome c	$\Delta\Psi_m$, cytochrome c, ROS	Separates stress signaling from organellar contamination	Supported ^{8,71,84}
Apoptosis	Caspase activation and apoptotic blebbing	caspase, annexin V/PI, ROCK1 blebbing	Determines apoptotic-body contribution	Established ^{20,83,121}
Necrosis/lysis	Irreversible membrane failure	LDH release, PI uptake, debris microscopy	Indicates unacceptable product contamination	Established ^{20,83}
Non-EV contaminants	Protein aggregates, organelle fragments, apoptotic bodies	calnexin, cytochrome c, albumin/lipoprotein markers	Defines purity and interpretability of EV readouts	Established ^{16,20,137,138}
Quality-control marker panel	Positive and negative EV markers	CD63/CD81/CD9, TSG101/ALIX/syntenin, negative markers	Connects model to MISEV-style QC	Established ^{41,48,138}

Abbreviations: EV, extracellular vesicle; mPTP, mitochondrial permeability transition pore; PI, propidium iodide; QC, quality control; MISEV, Minimal Information for Studies of Extracellular Vesicles.

$$\frac{dN_p}{dt} = k_{form}(\Delta V) - k_{reseal}N_p, \quad (A11)$$

$$J_{i,pore} = P_i(t)\Phi_i([i]_e, [i]_i, V_m), \quad (A12)$$

Table A14. Computational framework table for injury, debris, and quality gating.

Submodule	Model form	Representative relation	Inputs	Outputs
Damage accumulation	Damage-state ODE ^{1,12,84}	$\frac{dD}{dt} = k_{inj}G_m + k_{ROS}ROS - k_{repair}A_{repair}D$	conductance, repair	ROS, damage burden
Fate transition	Hybrid state rule ^{1,10,83}	$F_{apop} = H(D - D^*)$ or $\frac{dF_{apop}}{dt} = k_{apop}H(D - D^*)$	damage, mitochondria	ATP, viable/stressed/apoptotic fractions
Necrotic transition	Threshold rule ^{20,83}	$F_{nec} = H(D - D_{nec}^*)H(A_{ATP}^* - ATP)$	damage, membrane failure	ATP, necrotic fraction
Measured particles	Mixture model ^{16,20,137}	$N_{meas} = N_{EV} + N_{AB} + N_{debris} + N_{agg}$	release rates, injury state	measured particle count
Purity score	Ratio or classifier ^{16,137,138}	$Q_{EV} = \frac{N_{bona\ fide\ EV}}{N_{meas}}$	marker panel, debris state	EV quality score
Acceptable window	Constrained optimization ^{32,76,138}	maximize Y_{EV} subject to $Q_{EV} > Q^*$ and $Viability > V^*$	dose, cell state, harvest	operating region

Abbreviations: EV, extracellular vesicle; AB, apoptotic body; ATP, adenosine triphosphate; ROS, reactive oxygen species; QC, quality control.

Table A15. Biological/process evidence table for manufacturing, isolation, and quality-control interface.

Submodule	Process phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Harvest kinetics	Time-dependent EV accumulation	minute-to-hour release curves after nsPEF	Determines optimal collection window	Supported ^{32,64,76}
Cell-normalized yield	EV output per cell or protein	NTA/flow/ELISA normalized to cell number	Defines productivity	Established ^{64,79,138}
Isolation recovery	SEC, ultracentrifugation, density gradient, affinity capture	recovery and purity comparisons	Converts released EVs into isolated product	Established ^{16,136,137}
QC characterization	Size, morphology, markers, contaminants	NTA, cryo-EM/TEM, western blot/flow, proteomics	Determines whether product meets EV criteria	Established ^{41,48,138}
Batch variability	Donor, passage, culture, and pulse variability	repeated-batch studies, mixed-effects data	Determines reproducibility and manufacturability	Supported ^{64,79,119,136}
Scalable flow processing	Closed-system pulse chamber and inline monitoring	field uniformity, sterility, temperature, flow control	Enables translational manufacturing	Hypothesized ^{13,30,56,119}

Abbreviations: EV, extracellular vesicle; nsPEF, nanosecond pulsed electric field; NTA, nanoparticle tracking analysis; ELISA, enzyme-linked immunosorbent assay; QC, quality control; TEM, transmission electron microscopy.

Table A16. Computational framework table for manufacturing, isolation, and quality-control interface.

Submodule	Model form	Representative relation	Inputs	Outputs
Cell-normalized yield	Productivity metric ^{64,79,138}	$Y_{EV} = \frac{N_{EV}}{N_{cells}}$	EV count, cell number	EV yield per cell
Harvest accumulation	Time-integral model ^{32,64,76}	$N_{EV}(t_h) = \int_0^{t_h} R_{EV}(t)dt$	release rate, harvest time	collected EVs
Isolation recovery	Process-yield balance ^{16,136,137}	$N_{iso} = \eta_{iso}N_{released}$	isolation method, recovery	isolated EV count
Purity metric	Product-quality ratio ^{16,137,138}	$Purity = \frac{N_{EV}}{Protein_{total}}$ or marker-based score	particle/protein/marker data	purity estimate
Batch model	Mixed-effects model ^{64,79,119,136}	$Y_b = \mu + \alpha_{cell} + \beta_{dose} + u_b + \epsilon$	cell state, dose, batch	reproducibility
Optimization objective	Constrained optimization ^{64,79,119,136,138}	maximize $Potency \times Y_{iso}$ subject to purity and viability constraints	dose, harvest, isolation	optimal protocol

Abbreviations: EV, extracellular vesicle; QC, quality control.

$$\frac{d[Ca^{2+}]_i}{dt} = J_{Ca,pore} + J_{VGCC} + J_{ER,rel} - J_{SERCA} - J_{PMCA} - J_{NCX} - J_{mito} - J_{buffer}, \quad (A13)$$

$$A_x = \frac{[Ca^{2+}]_{sub}^n}{K_x^n + [Ca^{2+}]_{sub}^n}, \quad (A14)$$

$$R_{EV} = R_{sEV} + R_{m/LEV} + R_{AB}, \quad (A15)$$

$$R_{sEV} = k_{fus}f_{Ca}([Ca^{2+}]_{sub})M_d, \quad (A16)$$

$$R_{m/LEV} = k_{scission}f_{PS}(PS_{out})f_{actomyosin}B_{m/LEV}, \quad (A17)$$

$$Cargo_j = w_{j,sEV}R_{sEV} + w_{j,m/LEV}R_{m/LEV} + w_{j,AB}R_{AB}, \quad (A18)$$

$$\frac{dD}{dt} = k_{inj}G_m + k_{ROS}ROS - k_{repair}A_{repair}D, \quad (A19)$$

$$N_{meas} = N_{EV} + N_{AB} + N_{debris} + N_{agg}, \quad (A20)$$

$$\mathcal{V}_{EV} = \{N_{EV}, R_{EV}, Subtype, Cargo, Potency, Purity, Viability\} \quad (A21)$$

Variable and Parameter Glossary Template

Table A17 provides a shared notation layer for the variables and parameters introduced across the appendix tables and governing-equation scaffold.

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Table A17. Variable and parameter glossary for the nsPEF-EV engineering model.

Symbol	Definition	Units	Module	Source/estimation strategy
$E(t)$	Applied electric-field waveform	V/m	Dosimetry	Pulse protocol/field model
D_E	Electrical dose index	variable	Dosimetry	Derived from pulse protocol
ΔV_m	Plasma-membrane transmembrane potential	V	Electrodynamics	Circuit/electroporation model
ΔV_o	Organelle membrane potential perturbation	V	Organelle electrodynamics	Multi-shell/circuit model
N_p	Pore density or pore number	pores/ μm^2 or count	Electroporation	Fit or literature prior
r_p	Representative pore radius	nm	Electroporation	Literature prior or fit
G_m	Effective membrane conductance	S/m ²	Membrane/repair	Fit from permeability recovery
P_i	Ion-specific membrane permeability	m/s	Ion transport	Estimated from flux data
$J_{\text{Ca,pore}}$	Electropore-mediated Ca ²⁺ flux	concentration/time	Ion transport	Derived from permeability relation
$[\text{Ca}^{2+}]_i$	Cytosolic Ca ²⁺ concentration	M	Ca ²⁺ signaling	Ca ²⁺ imaging/model state
$[\text{Ca}^{2+}]_{\text{sub}}$	Submembrane Ca ²⁺ concentration	M	Microdomain	Reaction–diffusion/shell model
ROS	Reactive oxygen species state	variable	Stress/bioenergetics	ROS imaging/probes
ATP	Energy availability state	variable	Bioenergetics	ATP assays/model state
PS_{out}	Externalized phosphatidylserine fraction	dimensionless	Remodeling	Annexin V or model state
M_d	Docked MVB pool	count or concentration	EV release	Fit/estimated latent state
$B_{m/LEV}$	Medium/large EV budding precursor pool	count or concentration	EV release	Fit/estimated latent state
R_{sEV}	Small-EV release rate	particles/time	EV output	NTA/flow/marker-based fit
$R_{m/LEV}$	Medium/large EV release rate	particles/time	EV output	NTA/flow/marker-based fit
R_{AB}	Apoptotic body release rate	particles/time	Injury output	Apoptosis-weighted output
$Cargo_j$	Abundance of cargo class j	variable	Cargo	Proteomics/RNA/lipidomics
$Potency$	Functional activity metric	assay-specific	Potency	Recipient-cell assay
D	Integrated damage burden	dimensionless	Quality gate	Hybrid state variable
Q_{EV}	EV quality/purity score	dimensionless	QC	Marker/purity model
η_{iso}	Isolation recovery efficiency	dimensionless	Manufacturing	Process validation

Abbreviations: nsPEF, nanosecond pulsed electric field; EV, extracellular vesicle; ROS, reactive oxygen species; ATP, adenosine triphosphate; PS, phosphatidylserine; MVB, multivesicular body; ILV, intraluminal vesicle; QC, quality control.

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